

Feldman–Hájek, but Non-Gaussian: Geometry of the Weighted Fourier Jet

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Abstract

Feldman–Hájek theory gives an exact measure-class classification for Gaussian measures, but what replaces it for a non-Gaussian product law under arbitrary linear mixing? Kakutani’s criterion decides equivalence of infinite product measures, but it breaks down under bounded invertible linear mixing. We give one consolidated, complete answer and it is rigid.

1 Introduction

Let ρ be a non-Gaussian probability law and $\mu = \rho^{\otimes \mathbb{N}}$. We ask which bounded invertible operators on ℓ^2 preserve the measure class of μ when their rows act on the coordinate sequence. Kakutani’s theorem answers the corresponding question for coordinatewise transformations, but general linear mixing destroys the independence on which its product criterion relies. Fixed finite output marginals do not resolve the problem: they may all have positive densities even when the full image law is singular.

For finite-variance coordinate laws satisfying the regularity assumptions below, the answer has the same operator-ideal form as the Feldman–Hájek criterion, but with a discrete non-Gaussian stabilizer. The image law is equivalent to μ precisely when the mixing operator is a Hilbert–Schmidt perturbation of an allowed signed coordinate permutation; equivalently, the Hellinger energies of its finite output marginals are uniformly bounded in the number of output coordinates.

Stable laws and coordinate densities with zeros show how tail and positivity-set geometry alter this criterion in different ways.

1.1 Main theorem

Let $X \sim \rho$. Suppose $\rho(dx) = f(x) dx$, put $g = \sqrt{f}$, and assume

$$\begin{aligned} f > 0 \quad \text{Lebesgue-almost everywhere,} \quad g &\in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}X &= 0, \quad \mathbb{E}X^2 = 1, \quad f \neq (2\pi)^{-1/2}e^{-x^2/2}. \end{aligned} \tag{1}$$

Derivatives are weak; these hypotheses give finite location and scale Fisher information without requiring moments above order two.

Put $H = \ell^2(\mathbb{N}; \mathbb{R})$, let $(e_i)_{i \geq 1}$ be its coordinate basis, and write $\mathcal{S}_2(H)$ for the Hilbert–Schmidt ideal [35]. Define

$$\begin{aligned} \mathcal{P} = \{Q : (Qx)_i &= \varepsilon_i x_{\pi(i)}, \quad \pi \text{ a permutation of } \mathbb{N}, \\ \varepsilon_i &\in \{-1, 1\}, \quad \text{Law}(\varepsilon_i X) = \rho \text{ for every } i\}. \end{aligned} \tag{2}$$

Thus $\mathcal{P}$ is the full signed-permutation group when ρ is symmetric and the ordinary permutation group otherwise. For $T \in \mathcal{B}(H)$, let Φ_T be the measurable row-series realization constructed in lemma 3.1, and set $\nu_T = (\Phi_T)_\# \mu$. A sample from μ is not assumed to lie in H .

For probability measures λ, ν and any common dominating measure m , write

$$A(\lambda, \nu) = \int \sqrt{\frac{d\lambda}{dm} \frac{d\nu}{dm}} dm, \quad h(\lambda, \nu)^2 = 2 - 2A(\lambda, \nu). \quad (3)$$

Let $\mu_n = \rho^{\otimes n}$, let π_n extract the first n coordinates, and define

$$E_n(T) = -2 \log A(\mu_n, (\pi_n)_\# \nu_T). \quad (4)$$

These energies use the first n output rows of T , not its principal $n \times n$ corner.

Theorem 1.1 (Bounded invertible linear mixing). *Under (1), every $T \in \text{GL}(H)$ satisfies*

$$\boxed{\begin{aligned} \nu_T \sim \mu &\iff \exists Q \in \mathcal{P} : TQ^{-1} - I \in \mathcal{S}_2(H) \\ &\iff \sup_n E_n(T) < \infty. \end{aligned}} \quad (5)$$

If these conditions fail, then $\nu_T \perp \mu$ and $E_n(T) \uparrow \infty$. Moreover, $\nu_T = \mu$ if and only if $T \in \mathcal{P}$.

Bounded projected energy first detects a nonzero absolutely continuous component; the equivalence–singularity dichotomy upgrades that overlap to equivalence.

Corollary 1.2 (Moment-only stabilizer). *If ρ is non-Gaussian, centered, and has variance one, then the exact linear stabilizer of $\rho^{\otimes \mathbb{N}}$ is $\mathcal{P}$. No density or Fisher information assumption is needed.*

Corollary 1.3 (Orthogonal transformations). *If $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and f is non-Gaussian, centered, and has variance one, then every orthogonal U satisfies*

$$\nu_U \sim \mu \iff U - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}.$$

Thus scale Fisher information is not needed for orthogonal mixing.

1.2 Relation to earlier criteria

Kakutani characterizes equivalence of products with equivalent factors by summability of one-coordinate Hellinger losses [17]; for translations, Shepp relates this sum to square-root-density regularity [32]. Euclidean and Lie-group perturbations identify Fisher information as the local metric while retaining product structure [24, 37]. General output rows instead require estimates sensitive to the dependence created by mixing.

For Gaussian measures, Cameron–Martin describes admissible translations [7], while Feldman–Hájek theory and likelihood formulas describe equivalence under linear transformations [6, 10, 12, 33]. Ramer gives sufficient quasi-invariance and change-of-variables conditions for nonlinear Gaussian transformations [28]. Every orthogonal operator is an exact symmetry of the standard Gaussian product; in the non-Gaussian case the stabilizer is discrete, and equivalence is determined by Hilbert–Schmidt proximity to that stabilizer rather than by the covariance defect alone.

Shimomura's Theorem 3.4 shows that $I - S^2$ is compact for a self-adjoint admissible operator S , under ℓ^2 -continuity, ℓ^2 -translation quasi-invariance, and ℓ^2 -ergodicity; his examples preclude a universal Hilbert–Schmidt criterion under those assumptions alone [34]. Under (1), [theorem 1.1](#) gives the exact Hilbert–Schmidt condition modulo $\mathcal{P}$ for every bounded invertible operator. Other works construct non-Gaussian quasi-invariant measures on linear spaces [11, 38]; here we fix one i.i.d. product and classify its admissible linear maps.

Classical independence-of-linear-forms theorems [9, 15, 16, 36] characterize Gaussian laws through independence of suitable linear forms, while independent component analysis identifies non-Gaussian mixing matrices up to scaling and permutation [8]. Here the output coordinates may be dependent; their finite marginals instead detect square-summable deviation from an allowed signed coordinate permutation.

1.3 Two boundary regimes

For the standard symmetric α -stable product, $0 < \alpha < 2$, [theorem 4.1](#) shows, within the compatible stable row-series class, that equivalence holds precisely when an allowed signed coordinate permutation reduces the operator to a matrix $M = (m_{ij})$ satisfying

$$\sum_i |m_{ii} - 1|^2 + \sum_j \left(\sum_{i \neq j} |m_{ij}|^2 \right)^{\alpha/2} < \infty.$$

The off-diagonal term is an $\ell^\alpha(\ell^2)$ column cost, so positive density and finite Fisher information do not imply Hilbert–Schmidt sufficiency.

When a finite-variance density may vanish, Hilbert–Schmidt necessity persists, but [theorem 5.2](#) additionally requires every row sum of both T and T^{-1} to lie almost surely in $\{f > 0\}$. For bounded or one-sided coordinate laws, [theorem 5.3](#) reduces the equivalence group to explicit signed-permutation or permutation classes without density or Fisher assumptions.

1.4 Proof strategy

The finite-variance proof has three main steps. First, non-Gaussianity is encoded by the rotational Fourier defect

$$R(\xi, \eta) = \eta \phi'(\xi) \phi(\eta) - \xi \phi(\xi) \phi'(\eta), \quad \phi(u) = \mathbb{E} e^{iuX}.$$

The determinant of the two weighted Fourier jets $(\xi \phi(\xi), \phi'(\xi))$ and $(\eta \phi(\eta), \phi'(\eta))$ vanishes at every pair only for a Gaussian law. A finite set of frequencies then yields bounded tests for all matrix directions. Combined with the matrix Fisher operator, these tests give local Hellinger coercivity in Frobenius norm with constants independent of the dimension.

Second, the operator action must be realized on $\mathbb{R}^{\mathbb{N}}$, since a product sample need not belong to ℓ^2 . A common measurable carrier makes row series composable and allows inverse maps and likelihood cocycles to be used without incompatible null sets. Translation ergodicity gives the equivalence–singularity dichotomy and identifies bounded projected energy with equivalence.

Third, absolute continuity initially gives only asymptotic matching of remote rows with allowed signed coordinates. If the squared matching errors were not summable, tail-coordinate swaps would package a fixed amount of error into a small Hilbert–Schmidt perturbation. Local coercivity detects this perturbation, while absolute continuity makes the same remote swap asymptotically invisible. The contradiction proves Hilbert–Schmidt necessity.

We use A^* for the adjoint, P_n for coordinate projection on H , and $\|\cdot\|_F$ for the Frobenius norm.

2 Dimension-Free Hellinger Geometry

The rotational Fourier defect detects non-Gaussianity without dividing by the characteristic function. Fixed frequencies and the matrix Fisher operator then yield the dimension-free local estimate in [proposition 2.6](#); globalization is deferred to [Section 3.6](#).

2.1 A product of centered complex exponentials

Here only $\mathbb{E}X = 0$ and $\mathbb{E}|X| < \infty$ are required. Write $\phi(u) = \mathbb{E}e^{iuX}$, and for independent copies X_1, X_2 put

$$O_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad W_{\xi, \eta}(x, y) = (e^{i\xi x} - \phi(\xi))(e^{i\eta y} - \phi(\eta)).$$

With this sign convention, centering cancels the two marginal responses and isolates the dependence created by rotation. In particular,

$$\mathbb{E}W_{\xi, \eta}(X_1, X_2) = 0, \quad \mathbb{E}|W_{\xi, \eta}(X_1, X_2)|^2 = (1 - |\phi(\xi)|^2)(1 - |\phi(\eta)|^2) \leq 1. \quad (6)$$

Pointwise, $|W_{\xi, \eta}| \leq 4$, so its variance under an arbitrary comparison law is at most 16. Both bounds are dimension-free and require no tail assumption.

2.2 Gaussian characterization by a global linear ODE

The following logarithm-free rigidity argument remains valid when ϕ has zeros and is the infinitesimal counterpart of classical Gaussian independence characterizations [\[15, 16\]](#).

Lemma 2.1 (Gaussian characterization by the rotational Fourier defect). *For the product of centered complex exponentials defined above,*

$$R(\xi, \eta) = \left. \frac{d}{d\theta} \mathbb{E}W_{\xi, \eta}(O_\theta(X_1, X_2)) \right|_{\theta=0} = \eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta). \quad (7)$$

Moreover,

$$R \equiv 0 \iff \text{Law}(X) \text{ is a centered Gaussian, possibly degenerate.} \quad (8)$$

Proof. Since $\mathbb{E}|X| < \infty$, we have $\phi \in C^1(\mathbb{R})$ and $\phi'(0) = i\mathbb{E}X = 0$. After rotation, the joint characteristic function is

$$\phi(\xi \cos \theta + \eta \sin \theta)\phi(-\xi \sin \theta + \eta \cos \theta).$$

The two marginal terms in the centered product have zero derivative at $\theta = 0$, again by $\phi'(0) = 0$; differentiating the displayed product gives [\(7\)](#).

Now suppose $R \equiv 0$. Choose $\eta_0 \neq 0$ with $\phi(\eta_0) \neq 0$, possible because $\phi(0) = 1$. Then, for every ξ ,

$$\phi'(\xi) = c\xi\phi(\xi), \quad c = \frac{\phi'(\eta_0)}{\eta_0\phi(\eta_0)}, \quad \phi(0) = 1.$$

Uniqueness for this global linear ODE yields $\phi(\xi) = e^{c\xi^2/2}$. The identities $\phi(-\xi) = \overline{\phi(\xi)}$ and $|\phi(\xi)| \leq 1$ imply respectively $c \in \mathbb{R}$ and $c \leq 0$. Thus $c = -\sigma^2$ for some $\sigma \geq 0$, so X is centered Gaussian (the case $\sigma = 0$ is the point mass at zero). Conversely, direct substitution shows that every such Gaussian has $R \equiv 0$. $\square$

2.3 The weighted Fourier jet

The weighted Fourier jet

$$u \mapsto (u\phi(u), \phi'(u)) \in \mathbb{C}^2$$

satisfies

$$\det \begin{pmatrix} \xi\phi(\xi) & \phi'(\xi) \\ \eta\phi(\eta) & \phi'(\eta) \end{pmatrix} = -R(\xi, \eta). \quad (9)$$

Thus [lemma 2.1](#) says that a non-Gaussian law is detected by one pair of distinct frequencies. The factor u is essential: for a nondegenerate Gaussian law, the unweighted curve $u \mapsto (\phi(u), \phi'(u))$ is not contained in a fixed line.

2.4 Affine normalization

We record how R changes under affine normalization. Let Y have mean m and characteristic function ϕ_Y , and put $X = (Y - m)/\sigma$, where $\sigma > 0$. For $a = \xi/\sigma$ and $b = \eta/\sigma$, the rotational Fourier defect of X satisfies

$$R(\xi, \eta) = e^{-i(m/\sigma)(\xi+\eta)} [b\phi_Y'(a)\phi_Y(b) - a\phi_Y(a)\phi_Y'(b) + im(a-b)\phi_Y(a)\phi_Y(b)]. \quad (10)$$

The bracket is the mean-corrected response for two copies of Y : centering contributes the phase and scaling changes the frequencies. For a centered symmetric law the defect is real, but its phase at one pair is not a complete measure of skewness.

2.5 The matrix Fisher operator

The matrix Fisher operator resolves diagonal scale, symmetric off-diagonal, and rotational directions; this is the linear-mixing instance of the Fisher metric viewpoint [\[1\]](#).

Let f be a probability density and write $g = \sqrt{f}$. In this subsection we assume

$$\begin{aligned} f > 0 \quad \text{Lebesgue-almost everywhere,} \quad g &\in W^{1,2}(\mathbb{R}), \quad xg'(x) \in L^2(\mathbb{R}), \\ \mathbb{E}X &= 0, \quad \mathbb{E}X^2 = 1, \quad X \sim f(x) dx. \end{aligned} \quad (11)$$

The derivatives in these assumptions are weak derivatives. Define the location score $s = -f'/f = -2g'/g$ almost everywhere. Then

$$\mathbb{E}s(X) = 0, \quad \mathbb{E}[Xs(X)] = 1, \quad J := \mathbb{E}s(X)^2 = 4\|g'\|_2^2. \quad (12)$$

Define the scale score and its second moment by

$$d(x) := xs(x) - 1, \quad \kappa := \mathbb{E}d(X)^2 = 4\|xg'\|_2^2 - 1. \quad (13)$$

Here $f', xg' \in L^1$, so cutoff integration by parts against 1 and x gives the two identities in [\(12\)](#).

For independent copies $X_1, \dots, X_n$ of X , write $\mathbf{X}^{(n)} = (X_1, \dots, X_n) \sim \mu_n$. For $A \in M_n(\mathbb{R})$, define the score of the pushforward by $I + tA$ at $t = 0$ by

$$S_A(\mathbf{X}^{(n)}) = \sum_{i,j=1}^n A_{ij}(s(X_i)X_j - \delta_{ij}) = \sum_i A_{ii}d(X_i) + \sum_{i \neq j} A_{ij}s(X_i)X_j. \quad (14)$$

Theorem 2.2 (The matrix Fisher operator). *Under (11), let s and d be the location and scale scores defined above, and define the matrix Fisher operator $\mathcal{I}$ by $\langle A, \mathcal{I}B \rangle_F = \mathbb{E}[S_A S_B]$. Then*

$$\boxed{\mathcal{I}A = JA + A^* + (\kappa - J - 1) \text{diag } A.} \quad (15)$$

Here $\text{diag } A$ is the matrix with diagonal entries A_{ii} and zero off-diagonal entries. The diagonal, symmetric off-diagonal, and skew-symmetric matrix subspaces are orthogonal for the Frobenius inner product and have eigenvalues κ , $J + 1$, and $J - 1$, respectively. Equivalently,

$$\begin{aligned} \langle A, \mathcal{I}A \rangle_F &= \kappa \|\text{diag } A\|_F^2 + (J + 1) \left\| \frac{A + A^*}{2} - \text{diag } A \right\|_F^2 \\ &\quad + (J - 1) \left\| \frac{A - A^*}{2} \right\|_F^2. \end{aligned} \quad (16)$$

Their multiplicities are $n, n(n - 1)/2, n(n - 1)/2$, respectively. Moreover, $\kappa > 0$ and $J \geq 1$. Here $J = 1$ holds precisely for the standard Gaussian density. Thus, for a non-Gaussian f , $\min\{\kappa, J - 1\} > 0$ and

$$\begin{aligned} \min\{\kappa, J - 1\} \|A\|_F^2 &\leq \langle A, \mathcal{I}A \rangle_F \leq C_f \|A\|_F^2, \\ C_f &:= \max\{\kappa, J + 1\}. \end{aligned} \quad (17)$$

Proof. The variables $d(X_i)$ are centered and independent. Their covariances with the off-diagonal terms vanish, while two off-diagonal terms have nonzero covariance only when their ordered index pairs coincide or are reversed; the corresponding covariances are J and $(\mathbb{E}Xs(X))^2 = 1$. Hence

$$\mathbb{E}S_A^2 = \kappa \sum_i A_{ii}^2 + \sum_{i < j} \{J(A_{ij}^2 + A_{ji}^2) + 2A_{ij}A_{ji}\}. \quad (18)$$

Polarization gives (15); resolving each off-diagonal pair into its symmetric and skew-symmetric parts gives (16) and the stated multiplicities.

Cauchy–Schwarz applied to $\mathbb{E}Xs(X) = 1$ gives $J \geq 1$. Equality implies $s(X) = X$ almost surely, hence $g' = -xg/2$ almost everywhere and f is the standard Gaussian. If $\kappa = 0$, then $2xg' + g = 0$ almost everywhere on each of the two half-lines. The absolutely continuous solutions there are constant multiples of $|x|^{-1/2}$, which cannot give a positive probability density. This proves $\kappa > 0$ and the asserted bounds. $\square$

2.6 Hellinger paths, polar decomposition, and symmetric perturbations

In the left polar decomposition $T = BO$, covariance detects $B^2 - I = TT^* - I$, while the skew tangent of O has Fisher eigenvalue $J - 1$. The half-density formulation below packages both directions and is the square-root form of quadratic-mean differentiability [39, Section 7.2]; see also [3, 5] for differentiation of measures and quasi-invariance.

For $A \in \text{GL}(n, \mathbb{R})$ define a unitary operator on $L^2(\mathbb{R}^n, dx)$ by

$$(U_A v)(y) = |\det A|^{-1/2} v(A^{-1}y), \quad \Omega_n(x) = \prod_{i=1}^n g(x_i).$$

Thus $U_A \Omega_n$ is the square root of the Lebesgue density of $A_{\#} \mu_n$. For invertible A, B ,

$$h(A_{\#} \mu_n, B_{\#} \mu_n) = \|U_A \Omega_n - U_B \Omega_n\|_2.$$

For a fixed real matrix A , the generator of $U_{\exp(tA)}$ is

$$G_A v = -\frac{1}{2} \text{tr}(A) v - (Ax) \cdot \nabla v, \quad G_A \Omega_n = \frac{1}{2} S_A \Omega_n. \quad (19)$$

These identities hold in L^2 : the diagonal terms use $xg' \in L^2$, the off-diagonal terms use $g' \in L^2$ and $\mathbb{E}X^2 = 1$, and cutoff followed by mollification gives convergence in the required weighted Sobolev graph norm.

Lemma 2.3 (A dimension-free Hellinger path bound). *If A_t , $0 \leq t \leq 1$, is a continuously differentiable path in $\text{GL}(n, \mathbb{R})$, then $t \mapsto U_{A_t} \Omega_n$ is continuously differentiable in L^2 and*

$$\begin{aligned} \frac{d}{dt} U_{A_t} \Omega_n &= U_{A_t} G_{A_t^{-1} \dot{A}_t} \Omega_n, \\ \left\| \frac{d}{dt} U_{A_t} \Omega_n \right\|_2 &= \frac{1}{2} \sqrt{\left\langle A_t^{-1} \dot{A}_t, \mathcal{I}(A_t^{-1} \dot{A}_t) \right\rangle_F}. \end{aligned} \quad (20)$$

Consequently,

$$h((A_1)_{\#} \mu_n, (A_0)_{\#} \mu_n) \leq \frac{\sqrt{C_f}}{2} \int_0^1 \|A_t^{-1} \dot{A}_t\|_F dt. \quad (21)$$

In particular, if $\|B\|_{\text{op}} \leq r < 1$, then

$$h((I + B)_{\#} \mu_n, \mu_n) \leq \frac{\sqrt{C_f}}{2(1-r)} \|B\|_F. \quad (22)$$

All constants are independent of n .

Proof. The identity $U_A U_B = U_{AB}$ and (19) give the derivative formula. Unitarity and (17) give its norm; integration proves (21). For the last bound take $A_t = I + tB$ and use $\|A_t^{-1}\|_{\text{op}} \leq (1-r)^{-1}$. $\square$

2.7 Bounded Fourier tests for matrix tangents

Assume now that f is non-Gaussian. Although S_A is the tangent, using it as a moved-law test would require unavailable regularity and integrability. The rank-two obstruction in lemma 2.1 instead gives bounded Fourier tests for the off-diagonal directions; a small-frequency argument using the second moment supplies the diagonal test, all independently of n .

Lemma 2.4 (Finite real Fourier sums with dual moments). *There exist centered finite real Fourier sums p, q, r such that*

$$\begin{array}{c|cc} & s & x \\ \hline p & 1 & 0 \\ q & 0 & 1 \end{array} \quad \text{and} \quad \mathbb{E}[r(X)d(X)] = 1. \quad (23)$$

The entries in the table denote expectations of the products. For

$$H_A(\mathbf{X}^{(n)}) := \sum_i A_{ii} r(X_i) + \sum_{i \neq j} A_{ij} p(X_i) q(X_j), \quad (24)$$

one has, for all real matrices A, B of the same size,

$$\mathbb{E}[H_A S_B] = \langle A, B \rangle_F, \quad \|H_A\|_{L^2(\mu_n)} \leq C_{\text{dual}} \|A\|_F, \quad (25)$$

where C_{dual} is independent of n .

Proof. On the real space of centered finite real Fourier sums, consider

$$L_1(u) = \mathbb{E}u'(X) = \mathbb{E}[u(X)s(X)], \quad L_2(u) = \mathbb{E}[Xu(X)].$$

The integration-by-parts identity is valid because u, u' are bounded. If $u \mapsto (L_1(u), L_2(u))$ had rank at most one, then after complexification its determinant on the centered exponentials of frequencies ξ, η would vanish. Since $L_1(e^{i\xi x}) = i\xi\phi(\xi)$ and $L_2(e^{i\xi x}) = -i\phi'(\xi)$, this would give

$$\eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta) = 0 \quad (\xi, \eta \in \mathbb{R}).$$

By lemma 2.1, this contradicts non-Gaussianity. Thus the moment map has rank two, and inverting the moment matrix of two Fourier sums gives p, q with the required dual moments.

For $\xi \neq 0$ let $r_0(x) = \cos(\xi x) - \mathbb{E}\cos(\xi X)$. Dominated convergence gives

$$\frac{\mathbb{E}[Xr_0'(X)]}{\xi^2} = -\mathbb{E}\left[X\frac{\sin(\xi X)}{\xi}\right] \longrightarrow -\mathbb{E}X^2 = -1 \quad (\xi \rightarrow 0).$$

For small nonzero ξ the denominator is nonzero, so $r = r_0/\mathbb{E}[Xr_0'(X)]$ satisfies, by cutoff integration by parts, $\mathbb{E}[r(X)d(X)] = \mathbb{E}[Xr'(X)] = 1$.

Centering kills all diagonal–off-diagonal cross terms. Coincident ordered pairs contribute $\mathbb{E}[ps]\mathbb{E}[qX] = 1$, whereas reversed pairs contribute $\mathbb{E}[pX]\mathbb{E}[qs] = 0$; hence $\mathbb{E}[H_A S_B] = \langle A, B \rangle_F$. Finally, independence gives, for centered $a, b \in L^2(f)$,

$$\left\| \sum_{i \neq j} A_{ij} a(X_i) b(X_j) \right\|_2 \leq \sqrt{2} \|a\|_2 \|b\|_2 \|A\|_F. \quad (26)$$

Combining this with orthogonality of the centered diagonal terms proves the uniform norm bound. $\square$

To pair H_A with a moved Hellinger tangent, we need a derivative bound for $H_A \Omega_n$ uniform in n . Term counting loses dimension and direct Hölder estimates require unavailable moments; grouping index coincidences instead reduces every surviving coefficient array to a Frobenius contraction.

Lemma 2.5 (Generator bound for the functions H_A). *Let p, q, r be any fixed centered bounded continuously differentiable functions with bounded derivatives. Define H_A by (24). There is a constant C_{gen} , depending on these functions and on J, κ , such that*

$$\|G_B(H_A \Omega_n)\|_{L^2(dx)} \leq C_{\text{gen}} \|A\|_F \|B\|_F \quad (A, B \in M_n(\mathbb{R}), n \geq 1). \quad (27)$$

Proof. For fixed n , boundedness of H_A and its first derivatives, together with $\mathbb{E}X^2 = 1$ and the weighted Sobolev regularity of Ω_n , puts $H_A\Omega_n$ in the domain of G_B . The product rule gives

$$G_B(H_A\Omega_n) = \left\{ \frac{1}{2}H_AS_B - (Bx) \cdot \nabla H_A \right\} \Omega_n. \quad (28)$$

If $u_1, \dots, u_m$ are centered $L^2(f)$ functions and C is indexed by ordered tuples of *distinct* coordinates, then

$$\left\| \sum_{i_1, \dots, i_m}^{\neq} C_{i_1 \dots i_m} \prod_{a=1}^m u_a(X_{i_a}) \right\|_2 \leq \sqrt{m!} \|C\|_{\ell^2} \prod_{a=1}^m \|u_a\|_2. \quad (29)$$

After squaring, only pairs of tuples with the same index set survive; summing their at most $m!$ matchings proves the estimate. We use it for $m \leq 4$.

For a vector a and a zero-diagonal matrix M , abbreviate

$$D_a(u) = \sum_i a_i u(X_i), \quad Q_M(u, v) = \sum_{i \neq j} M_{ij} u(X_i) v(X_j).$$

Then H_AS_B is obtained by multiplying $D_{\mathbf{a}}(r) + Q_{A_0}(p, q)$ and $D_{\mathbf{b}}(d) + Q_{B_0}(s, x)$, where $\mathbf{a} = (A_{ii})_i$, $\mathbf{b} = (B_{ii})_i$, $A_0 = A - \text{diag } A$, and $B_0 = B - \text{diag } B$. Every local product on a shared coordinate is a bounded function times one of d, s, x , hence is in $L^2(f)$; split it into its mean and centered part. For $D_a D_b$, the distinct-index tensor, Hadamard vector, and scalar contraction have norm at most $\|a\|_2 \|b\|_2$. For $D_a Q_M$, the three coincidence classes yield $a \otimes M$, a row or column multiplication of M , and Ma or M^*a , all bounded by $\|a\|_2 \|M\|_F$.

For the remaining product

$$Q_M(u, v) Q_N(w, z) = \sum_{i \neq j, k \neq l} M_{ij} N_{kl} u_i v_j w_k z_l,$$

with u, v bounded and $w, z \in L^2$, four distinct indices give $M \otimes N$. In the representative one-coincidence pattern $i = k$, the centered coefficient has norm

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_l N_{il}^2 \right) \right\}^{1/2} \leq \|M\|_F \|N\|_F.$$

Its mean gives the off-diagonal part of M^*N ; the other one-coincidence classes are transposes. When both indices are shared, Cauchy–Schwarz bounds the Hadamard coefficients $M_{ij}N_{ij}$, $M_{ij}N_{ji}$, and all their row, column, and scalar contractions by $\|M\|_F \|N\|_F$. These are all partitions of the two ordered pairs. Applying (29) to the centered pieces proves

$$\|H_AS_B\|_2 \leq C \|A\|_F \|B\|_F. \quad (30)$$

For $(Bx) \cdot \nabla H_A$, the derivative of the diagonal part is

$$\sum_{i,k} A_{ii} B_{ik} r'(X_i) X_k.$$

After centering r' and xr' , its coefficient arrays are $(\text{diag } A)B$, $B^*\mathbf{a} - (a_i b_i)_i$, and a diagonal contraction; their norms are at most $2\|A\|_F \|B\|_F$.

With $M = A_0$, the off-diagonal derivative is the sum of

$$F(M, B; p', q) := \sum_{i \neq j, k} M_{ij} B_{ik} p'(X_i) q(X_j) X_k$$

and $F(M^*, B; q', p)$. For $k \notin \{i, j\}$, centering p' gives the three-index bound

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_k B_{ik}^2 \right) \right\}^{1/2} \leq \|M\|_F \|B\|_F,$$

and its mean gives $M^* B - M^* \text{diag } B$. The classes $k = i$ and $k = j$ give $(B_{ii} M_{ij})$, $M^* \mathbf{b}$, $(M_{ij} B_{ij})_{ij}$, and their row, column, and scalar contractions. The same Frobenius estimates control each array, while boundedness of p', q and $\mathbb{E} X^2 = 1$ puts every local factor in L^2 . Applying the same argument to $F(M^*, B; q', p)$ and then (29) yields

$$\|(Bx) \cdot \nabla H_A\|_2 \leq C \|A\|_F \|B\|_F.$$

Together with (30) and (28), this proves the lemma. $\square$

Pairing H_A with the tangent detects the first-order displacement; the generator bound controls the error along the exponential path.

Proposition 2.6 (Uniform local Hellinger coercivity). *For every non-Gaussian density satisfying (11), there are constants $c, C, \delta > 0$, independent of n , such that*

$$c \|B\|_F \leq h((I + B)_{\#} \mu_n, \mu_n) \leq C \|B\|_F \quad \text{whenever } \|B\|_F \leq \delta. \quad (31)$$

The lower bound can be obtained using the fixed finite set of coordinate frequencies from lemma 2.4.

Proof. Fix the finite real Fourier sums from lemma 2.4. In exponential coordinates, unitarity and the group derivative give

$$\langle H_A \Omega_n, (U_{\exp A} - I) \Omega_n \rangle = \int_0^1 \langle U_{\exp(-tA)} (H_A \Omega_n), G_A \Omega_n \rangle dt.$$

The value obtained by replacing $U_{\exp(-tA)} (H_A \Omega_n)$ with $H_A \Omega_n$ is $\frac{1}{2} \mathbb{E} H_A S_A = \frac{1}{2} \|A\|_F^2$. Moreover, $\|(U_{\exp(-tA)} - I)v\|_2 \leq t \|G_A v\|_2$ on the generator domain. Therefore lemma 2.5 and (17) yield

$$\left| \langle H_A \Omega_n, (U_{\exp A} - I) \Omega_n \rangle - \frac{1}{2} \|A\|_F^2 \right| \leq \frac{C_{\text{gen}} \sqrt{C_f}}{4} \|A\|_F^3.$$

For sufficiently small $\|A\|_F$, Cauchy–Schwarz and (25) thus give

$$h((\exp A)_{\#} \mu_n, \mu_n) \geq \frac{1}{4C_{\text{dual}}} \|A\|_F.$$

For $\|B\|_{\text{op}} < 1/2$, $A = \log(I + B)$ is real and $\|A - B\|_F \leq C \|B\|_{\text{op}} \|B\|_F$. Hence $\|A\|_F$ and $\|B\|_F$ are uniformly comparable in a fixed chart, proving the lower bound. The upper bound is lemma 2.3. $\square$

2.8 Uniform coercivity modulo the exact linear stabilizer

The following global form is proved in [Section 3.6](#), after the infinite-dimensional classification supplies the compactness input; it is not used in that classification.

Theorem 2.7 (Dimension-free global Hellinger coercivity). *Assume (1). Let $\mathcal{P}_n$ denote the allowed signed-permutation matrices in dimension n , and set*

$$d_n(T) = \min_{Q \in \mathcal{P}_n} \|T - Q\|_F. \quad (32)$$

For every $0 < a \leq b < \infty$ there are constants $c = c(f, a, b) > 0$ and $C = C(f, a, b) < \infty$ such that, for every n and every $T \in \text{GL}(n, \mathbb{R})$ whose singular values lie in $[a, b]$,

$$c \min\{d_n(T)^2, 1\} \leq h(\mu_n, T_{\#}\mu_n)^2 \leq C \min\{d_n(T)^2, 1\}. \quad (33)$$

3 The Finite-Variance Classification

After realizing row series on a common measurable carrier, we prove Hilbert–Schmidt sufficiency by approximation and necessity by combining tail rigidity with conjugated coordinate swaps.

3.1 Measurable row-series realizations on countable operator orbits

Only $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$ are needed here. Each row series exists as an L^2 equivalence class. The distinction between an operator on ℓ^2 and its measurable realization on $\mathbb{R}^{\mathbb{N}}$ is already central to [\[34\]](#). Such equivalence classes cannot be composed pointwise: separately chosen versions for different operators or orbit laws may fail on incompatible null sets. We therefore select deterministic subsequences for every row in a fixed countable operator group, so that operator multiplication becomes composition on one common measurable carrier. Countability is used only to intersect the exceptional sets; no universal pointwise action of all of $\text{GL}(H)$ is asserted.

Lemma 3.1 (Measurable row-series realization on each countable operator orbit). *For every $T \in \mathcal{B}(H)$, there is a measurable row-series realization, namely a Borel map $\Phi_T : \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}^{\mathbb{N}}$ whose coordinates are the $L^2(\mu)$ row sums*

$$(\Phi_T x)_i = \sum_{j \geq 1} T_{ij} x_j.$$

These maps can be chosen so that the following statements hold. If a probability measure λ satisfies, for some finite $C_\lambda \geq 0$,

$$\int \left| \sum_j a_j x_j \right|^2 d\lambda(x) \leq C_\lambda \|a\|_2^2 \quad \text{for every finitely supported } a, \quad (34)$$

then $(\Phi_T)_{\#}\lambda$ also satisfies this condition. Its constant is at most $C_\lambda \|T\|_{\text{op}}^2$. Also

$$\Phi_S \circ \Phi_T = \Phi_{ST} \quad \lambda\text{-almost surely} \quad (S, T \in \mathcal{B}(H)). \quad (35)$$

For each fixed countable subgroup $\Gamma \leq \text{GL}(H)$, there is a Γ -invariant Borel set $D \subset \mathbb{R}^{\mathbb{N}}$ of full measure for every λ satisfying (34). On this set, these restrictions form an action by bimeasurable automorphisms. In particular, this applies to μ and all its Γ -orbit laws, including singular orbit laws.

Proof. For $a \in \ell^2$, the finite sums $\sum_{j \leq n} a_j x_j$ have a unique $L^2(\lambda)$ limit L_a . Choose

$$n_k(a) = \min \left\{ n \geq k : \sum_{j > n} |a_j|^2 \leq 2^{-4k} \right\}.$$

Define $\hat{L}_a(x)$ to be the limit of $\sum_{j \leq n_k(a)} a_j x_j$ when it exists, and zero otherwise. By (34),

$$\left\| \sum_{j \leq n_k(a)} a_j x_j - L_a \right\|_{L^2(\lambda)}^2 \leq C \lambda 2^{-4k}.$$

Chebyshev's inequality and Borel–Cantelli give convergence along this subsequence λ -almost surely and in $L^2(\lambda)$. This deterministic choice works for every law satisfying (34). Set $\Phi_T(x)_i = \hat{L}_{T^*e_i}(x)$; the construction is Borel.

For finitely supported b , $\sum_i b_i \Phi_T(x)_i = L_{T^*b}$ almost surely, proving the asserted second-moment bound for the pushforward. For the i -th row of S ,

$$\sum_{j \leq n} S_{ij} \hat{L}_{T^*e_j} = L_{T^*P_n S^*e_i} \longrightarrow L_{T^*S^*e_i} \quad \text{in } L^2(\lambda).$$

The selected subsequence on the left also converges almost surely to $\Phi_S(\Phi_T(x))_i$, because the output law satisfies the same type of bound. Uniqueness of the limit proves (35).

For a countable Γ , let B be the Borel set on which all its row limits, identity relations, and pairwise composition relations hold. The preceding argument gives $\lambda(B) = 1$ for every admissible law. Put

$$D = \bigcap_{T \in \Gamma} \Phi_T^{-1}(B).$$

Every orbit law satisfies (34), so D has full measure. The composition identities make D invariant and $\Phi_{T^{-1}}$ the inverse of Φ_T on D . $\square$

Write $\nu_T = (\Phi_T)_\# \mu$.

3.2 Rational translation invariance and the equivalence–singularity dichotomy

The relevant translation theory identifies shifts preserving product and coordinatewise measure classes [2, 13, 14, 29, 32]. Here ℓ^2 -quasi-invariance and rational-translation ergodicity specialize the equivalence–singularity dichotomy of [27] to each bounded invertible orbit.

Lemma 3.2 (Equivalence–singularity dichotomy on every bounded invertible orbit). *For every $T \in \text{GL}(H)$,*

$$\nu_T \sim \mu \quad \text{or} \quad \nu_T \perp \mu. \tag{36}$$

This conclusion uses only the a.e. positivity of f , its centered finite second moment, and $\sqrt{f} \in W^{1,2}(\mathbb{R})$. In particular, it holds under (1).

Proof. Put $g = \sqrt{f}$. The translation estimate

$$1 - \mathbf{A}(f, f(\cdot - a)) = \frac{1}{2} \|g(\cdot - a) - g\|_2^2 \leq \frac{1}{2} \|g'\|_2^2 a^2$$

and Kakutani's theorem imply $(x \mapsto x + a)_{\#}\mu \sim \mu$ for every $a \in \ell^2$. Since the scalar pairing of a row of T with $T^{-1}a$ is absolutely convergent,

$$\Phi_T(x + T^{-1}a) = \Phi_T(x) + a \quad \mu\text{-almost surely};$$

hence ν_T is also quasi-invariant under every ℓ^2 translation.

Let $\mathcal{A}$ be the countable additive group generated by $\{qe_j, qTe_j : q \in \mathbb{Q}, j \geq 1\}$. Both μ and ν_T are quasi-invariant and ergodic under translation by $\mathcal{A}$. Indeed, rational-translation invariance and positivity of the finite-dimensional densities make every finite-coordinate section constant almost everywhere (convolve with a smooth approximate identity); the event is therefore a tail event, so Kolmogorov's zero-one law applies. Pullback by Φ_T proves the same assertion for ν_T , because translation by qTe_j pulls back to translation by qe_j .

Write $\nu_T = (\nu_T)_{ac} + (\nu_T)_s$ for its Lebesgue decomposition relative to μ . Choose a Borel μ -null set N_0 supporting $(\nu_T)_s$, and let $N = \bigcup_{a \in \mathcal{A}} (N_0 + a)$. Quasi-invariance of μ makes N a μ -null set; it is exactly $\mathcal{A}$ -invariant and still supports $(\nu_T)_s$. Thus $\nu_T(N) = (\nu_T)_s(\mathbb{R}^N) \in \{0, 1\}$ by ergodicity. Mass one gives singularity. If the mass is zero, write $\nu_T = \ell_T \mu$. Quasi-invariance makes $\{\ell_T > 0\}$ invariant modulo μ ; it has positive measure and hence, by ergodicity, full measure. Thus $\nu_T \sim \mu$. $\square$

3.3 Finite-marginal likelihood ratios and projected Hellinger energy

The finite-marginal martingale converges only to the density of the absolutely continuous part of ν_T ; the equivalence-singularity dichotomy upgrades positive limiting affinity to equivalence. See [19, 22] for the likelihood-ratio and Lebesgue-decomposition background, and [18] for the distinct problem of independent additive perturbations.

Lemma 3.3 (Projected Hellinger energy and finite-marginal likelihood ratios). *Assume $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$. For $T \in \text{GL}(H)$, every finite output marginal $(\pi_n)_{\#}\nu_T$ is equivalent to μ_n . Writing $\mathcal{F}_n = \sigma(x_1, \dots, x_n)$, define*

$$\ell_{T,n}(x) = \frac{d[(\pi_n)_{\#}\nu_T]}{d\mu_n}(\pi_n x). \quad (37)$$

Then $A(\mu_n, (\pi_n)_{\#}\nu_T) = \mathbb{E}_\mu \sqrt{\ell_{T,n}}$, and $E_n(T)$ is as in (4); moreover, $(\ell_{T,n})_{n \geq 1}$ is a nonnegative mean-one μ -martingale. We have

$$A(\mu_n, (\pi_n)_{\#}\nu_T) \downarrow A(\mu, \nu_T), \quad \nu_T \sim \mu \iff \sup_n E_n(T) < \infty. \quad (38)$$

Otherwise $E_n(T) \uparrow \infty$ and $\nu_T \perp \mu$. If $\nu_T \sim \mu$ with density $\ell_T = d\nu_T/d\mu$, then

$$\ell_{T,n} = \mathbb{E}_\mu[\ell_T \mid \mathcal{F}_n] \longrightarrow \ell_T \quad \text{in } L^1(\mu), \quad (39)$$

$$\ell_{T,n}^{-1} = \mathbb{E}_{\nu_T}[\ell_T^{-1} \mid \mathcal{F}_n] \longrightarrow \ell_T^{-1} \quad \text{in } L^1(\nu_T). \quad (40)$$

In particular, both likelihood-ratio martingales are uniformly integrable in their respective reference measures.

Proof. The first n rows of T are linearly independent, since T^* is injective, so some n input columns form an invertible minor A . Writing the first n outputs as $AU + W$, with $U \sim \mu_n$

independent of W , shows that their density is a mixture of translates of the a.e.-positive density of AU . It is positive and finite almost everywhere, proving finite-dimensional equivalence.

Consistency of projected laws proves the martingale assertion. The martingale limit is $\ell_{T,\infty} = d(\nu_T)_{ac}/d\mu$, the density of the absolutely continuous part of ν_T . Indeed, Fatou on cylinder sets gives $\ell_{T,\infty}\mu \leq \nu_T$; conversely, if $(\nu_T)_{ac} = \ell_{ac}\mu$, then $\mathbb{E}_\mu[\ell_{ac} \mid \mathcal{F}_n] \leq \ell_{T,n}$, and martingale convergence gives the reverse inequality. Since $\mathbb{E}_\mu(\sqrt{\ell_{T,n}})^2 = 1$, the square roots are uniformly integrable and converge in L^1 to $\sqrt{\ell_{T,\infty}}$. Conditional Jensen also makes their expectations decreasing; hence

$$\lim_n \mathbf{A}(\mu_n, (\pi_n)_\# \nu_T) = \int \sqrt{\ell_{T,\infty}} d\mu = \mathbf{A}(\mu, \nu_T).$$

The dichotomy in [lemma 3.2](#) now proves (38) and the singular alternative.

Under equivalence, the forward identity is the defining conditional-expectation identity for a marginal density. For the reverse, for $B \in \mathcal{F}_n$,

$$\int_B \ell_{T,n}^{-1} d\nu_T = \mu(B) = \int_B \ell_T^{-1} d\nu_T.$$

Conditional-expectation convergence gives both L^1 limits and hence both uniform-integrability assertions. $\square$

Since initial segments are cofinal among finite coordinate sets, Hellinger data processing also makes the criterion independent of their enumeration [21].

3.4 Sufficiency of Hilbert–Schmidt perturbations

Finite-coordinate Hilbert–Schmidt approximants are Cauchy in Hellinger distance, but this alone produces only a candidate density and one direction of absolute continuity. The inverse approximants and the common measurable carrier are what prevent the limit from losing mass in the reverse direction.

Proposition 3.4 (Hilbert–Schmidt sufficiency and likelihood-ratio convergence). *Assume (11) and let $T = I + K \in \text{GL}(H)$ with $K \in \mathcal{S}_2$. Then $\nu_T \sim \mu$. For the eventually invertible operators $T_n = I + P_n K P_n$, their likelihood ratios $\ell_{T_n} = d\nu_{T_n}/d\mu$ converge in $L^1(\mu)$ to the strictly positive likelihood ratio $\ell_T = d\nu_T/d\mu$, and $\sqrt{\ell_{T_n}} \rightarrow \sqrt{\ell_T}$ in $L^2(\mu)$. The inverse approximants likewise satisfy $\ell_{T_n^{-1}} \rightarrow \ell_{T^{-1}}$ in $L^1(\mu)$. If Q is an exact linear stabilizer of μ , then $\nu_T \sim \mu$ whenever $TQ^{-1} - I \in \mathcal{S}_2$ and T is invertible.*

Proof. Since $P_n K P_n \rightarrow K$ in Hilbert–Schmidt norm, $T_n \rightarrow T$ in operator norm, so the approximants are eventually invertible with uniformly bounded inverse norms; their finite-coordinate likelihood ratios ℓ_{T_n} are positive almost everywhere. For all sufficiently large m, n , the operator $T_n^{-1}T_m - I$ acts in a finite coordinate block and has operator norm at most $1/2$. Hellinger invariance under the finite invertible map T_n^{-1} and (22) give

$$\|\sqrt{\ell_{T_m}} - \sqrt{\ell_{T_n}}\|_2 \leq \sqrt{C_f} \|T_n^{-1}\|_{\text{op}} \|T_m - T_n\|_{\mathcal{S}_2}. \quad (41)$$

Thus $\sqrt{\ell_{T_n}}$ converges in $L^2(\mu)$ to a nonnegative function of norm one; denote its square by $\tilde{\ell}_T$. Since

$$\|\ell_{T_m} - \ell_{T_n}\|_1 \leq 2\|\sqrt{\ell_{T_m}} - \sqrt{\ell_{T_n}}\|_2,$$

we also have $\ell_{T_n} \rightarrow \tilde{\ell}_T$ in L^1 . Meanwhile, $\Phi_{T_n} \mathbf{X} \rightarrow \Phi_T \mathbf{X}$ in L^2 on every finite set of output coordinates. Cylinder tests therefore identify $\ell_T \mu$ with ν_T .

The inverse approximants converge in Hilbert–Schmidt norm because

$$\|T_n^{-1} - T^{-1}\|_{\mathcal{S}_2} \leq \|T_n^{-1}\|_{\text{op}} \|T_n - T\|_{\mathcal{S}_2} \|T^{-1}\|_{\text{op}}.$$

Applying the same argument gives $\nu_{T^{-1}} \ll \mu$. On the common measurable carrier supplied by lemma 3.1,

$$\mu = (\Phi_T)_\# \nu_{T^{-1}} \ll (\Phi_T)_\# \mu = \nu_T.$$

This proves equivalence and identifies $\tilde{\ell}_T = \ell_T > 0$ almost everywhere. Finally, $\nu_Q = \mu$ and $T = (TQ^{-1})Q$ give the assertion about exact stabilizers. $\square$

3.4.1 The orthogonal case requires only location Fisher information

For orthogonal transformations the assumptions can be reduced to

$$f > 0 \text{ almost everywhere, } g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1. \quad (42)$$

Thus $xg' \in L^2$ is not required. Cutoff integration by parts still gives $\mathbb{E}s(X) = 0$ and $\mathbb{E}[Xs(X)] = 1$, while a skew-symmetric A has only off-diagonal score terms and

$$\mathbb{E}S_A^2 = (J - 1)\|A\|_F^2. \quad (43)$$

Only $x_j \partial_i \Omega_n$ with $i \neq j$ occur in the generator formula, which follows by radial cutoff and mollification commuting with the orthogonal one-parameter group.

Proposition 3.5 (Orthogonal local bounds and Hilbert–Schmidt sufficiency). *Under (42), every orthogonal U on ℓ^2 with $U - I \in \mathcal{S}_2$ satisfies $\nu_U \sim \mu$. Its likelihood ratios have the same L^1 and square-root L^2 convergence along finite-coordinate orthogonal approximants, as do the inverse likelihood ratios. If f is non-Gaussian, there also exist dimension-free $c, C, \delta > 0$ such that*

$$c\|V - I\|_F \leq h(V_\# \mu_n, \mu_n) \leq C\|V - I\|_F \quad (V \in O(n), \|V - I\|_F \leq \delta). \quad (44)$$

Proof. For a skew-symmetric A , integration along $\exp(tA)$ gives

$$h((\exp A)_\# \mu_n, \mu_n) \leq \frac{1}{2} \sqrt{J - 1} \|A\|_F. \quad (45)$$

The construction of p, q in lemma 2.4 uses only $s(X), X \in L^2$ and $J - 1 > 0$. For skew-symmetric A , take $H_A = \sum_{i \neq j} A_{ij} p(X_i) q(X_j)$. Both H_A and S_A have no diagonal term, so the generator proof uses only bounded functions times $s(X)$ or X and remains valid under (42). The half-density pairing gives the local lower bound in exponential coordinates. Since a sufficiently small orthogonal $V - I$ has a real skew-symmetric logarithm with $\|\log V\|_F \asymp \|V - I\|_F$, this proves (44).

For the infinite-dimensional assertion, set $T_n = I + P_n(U - I)P_n$ and take its polar factor

$$V_n = T_n(T_n^* T_n)^{-1/2}.$$

It is orthogonal and acts identically outside the first n coordinates. Since $T_n \rightarrow U$ in Hilbert–Schmidt norm and its least singular value is eventually bounded below by some $a > 0$,

$$\|V_n - T_n\|_{\mathcal{S}_2} = \|I - |T_n|\|_{\mathcal{S}_2} \leq \frac{\|T_n^* T_n - I\|_{\mathcal{S}_2}}{1 + a}.$$

The right side tends to zero, so $V_n \rightarrow U$ in Hilbert–Schmidt norm. For large m, n , apply (45) to the small skew-symmetric logarithm of $V_n^* V_m$. The square roots of the likelihood ratios are Cauchy, and cylinder tests identify their limit as in [proposition 3.4](#). Repeating the argument for $V_n^{-1} \rightarrow U^{-1}$ and using the measurable inverse action proves equivalence and all asserted convergence. $\square$

3.5 Necessity of Hilbert–Schmidt perturbations

Lemma 3.6 (The local bound for Hilbert–Schmidt operators). *Let f be non-Gaussian and satisfy (11). There are constants $c, C, \delta > 0$ such that every $K \in \mathcal{S}_2(\ell^2)$ with $\|K\|_{\mathcal{S}_2} \leq \delta$ satisfies*

$$c\|K\|_{\mathcal{S}_2} \leq h(\nu_{I+K}, \mu) \leq C\|K\|_{\mathcal{S}_2}. \quad (46)$$

The constants can be chosen with $\delta < 1$, so $I + K$ is automatically invertible. Under the weaker assumptions (42), the same conclusion holds for orthogonal $I + K$ whenever f is non-Gaussian.

Proof. Let $K_n = P_n K P_n$. Since the untouched tail has affinity one, (31) applies to $I + K_n$, while [proposition 3.4](#) gives

$$h(\nu_{I+K_n}, \mu) \longrightarrow h(\nu_{I+K}, \mu), \quad \|K_n\|_{\mathcal{S}_2} \longrightarrow \|K\|_{\mathcal{S}_2}.$$

Passing to the limit proves both bounds.

For the orthogonal assertion, choose the radius in this lemma to be half the finite-dimensional radius in (44). The polar approximants from [proposition 3.5](#) satisfy

$$h(\nu_{V_n}, \mu) \longrightarrow h(\nu_{I+K}, \mu), \quad \|V_n - I\|_{\mathcal{S}_2} \longrightarrow \|K\|_{\mathcal{S}_2},$$

Thus all sufficiently large approximants lie within the finite-dimensional radius, and (44) passes to the limit. $\square$

3.5.1 Characteristic-function rigidity and bounded invertible linear mixing

The next two lemmas use only a centered, variance-one, non-Gaussian coordinate law: a row whose linear combination converges in law to X must approach an allowed signed coordinate.

Lemma 3.7 (A product approximation for small coefficients). *There is a nonnegative function ω_ρ , with $\omega_\rho(\delta) \rightarrow 0$ as $\delta \downarrow 0$, such that, for every finite or countable real family (b_j) with $\sum_j b_j^2 < \infty$,*

$$\left| \prod_j \phi(tb_j) - \exp\left(-\frac{t^2}{2} \sum_j b_j^2\right) \right| \leq \omega_\rho(\delta) t^2 \sum_j b_j^2 \quad \text{if} \quad \sup_j |tb_j| \leq \delta. \quad (47)$$

For a countable family, the product denotes the characteristic function of the L^2 sum $\sum_j b_j X_j$.

Proof. The centered second-moment assumption gives

$$\phi(u) = 1 - \frac{u^2}{2} + o(u^2).$$

Consequently

$$\omega_\rho(\delta) := \sup_{0 < |u| \leq \delta} \frac{|\phi(u) - e^{-u^2/2}|}{u^2} \longrightarrow 0, \quad \omega_\rho(0) := 0.$$

Both factors have modulus at most one. Telescoping finite products therefore bounds their difference by $\sum_j \omega_\rho(\delta) t^2 b_j^2$, proving (47). The countable case follows from L^2 convergence of the partial sums. $\square$

Lemma 3.8 (Qualitative rigidity of a unit row). *Let $a^{(n)} \in \ell^2$ satisfy $\|a^{(n)}\|_2 = 1$. If*

$$\text{Law}\left(\sum_j a_j^{(n)} X_j\right) \Rightarrow \rho,$$

then

$$\inf_{\substack{j \geq 1, \varsigma \in \{-1, 1\} \\ \text{Law}(\varsigma X) = \rho}} \|a^{(n)} - \varsigma e_j\|_2 \longrightarrow 0. \quad (48)$$

Proof. Reorder each row by decreasing absolute value; this preserves the law of the sum and gives $|a_j^{(n)}| \leq j^{-1/2}$. From any subsequence, extract a further one with $a_j^{(n)} \rightarrow a_j$ for every j . Fatou gives

$$\tau_a := \sum_j a_j^2 \leq 1.$$

For fixed m, t , lemma 3.7 replaces the tail characteristic function by

$$\exp\left[-\frac{t^2}{2} \left(1 - \sum_{j \leq m} (a_j^{(n)})^2\right)\right]$$

with error at most $t^2 \omega_\rho(|t|/\sqrt{m+1})$, uniformly in n . Letting first $n \rightarrow \infty$ and then $m \rightarrow \infty$ gives

$$e^{-(1-\tau_a)t^2/2} \prod_j \phi(a_j t). \quad (49)$$

Thus the limit is $\sum_j a_j X_j$ plus an independent $N(0, 1 - \tau_a)$ variable.

Under the convergence hypothesis this limiting law is the law of X . Suppose that no $|a_j|$ is one. If all a_j vanish, (49) already makes X Gaussian. Otherwise $m_a := \max_j |a_j|$ exists and lies in $(0, 1)$. For $\mathbf{j} = (j_1, \dots, j_k) \in \mathbb{N}^k$, set $b_{\mathbf{j}} := a_{j_1} \cdots a_{j_k}$. Iterating the characteristic-function identity (49) k times gives

$$\phi(t) = e^{-(1-\tau_a^k)t^2/2} \prod_{\mathbf{j} \in \mathbb{N}^k} \phi(t b_{\mathbf{j}}), \quad (50)$$

where

$$\sum_{\mathbf{j}} b_{\mathbf{j}}^2 = \tau_a^k, \quad \sup_{\mathbf{j}} |b_{\mathbf{j}}| \leq m_a^k.$$

The substitution is legitimate in L^2 at every finite depth. Applying lemma 3.7 to (50) yields

$$|\phi(t) - e^{-t^2/2}| \leq t^2 \tau_a^k \omega_\rho(|t| m_a^k) \longrightarrow 0.$$

This contradicts non-Gaussianity. Hence some $a_j = \varsigma \in \{-1, 1\}$. All other coefficients and the Gaussian summand vanish, and the limiting identity gives $\text{Law}(\varsigma X) = \rho$.

If (48) failed, a subsequence would stay a fixed distance from every allowed signed unit vector. The preceding extraction produces an allowed ςe_j with

$$\|a^{(n)} - \varsigma e_j\|_2^2 = 2 - 2\varsigma a_j^{(n)} \longrightarrow 0.$$

Undoing the permutations is a contradiction. $\square$

Lemma 3.9 (Uniform invariance under tail symmetries). *Let μ be the i.i.d. product law and let $\nu \ll \mu$. Let $\mathcal{T}_N$ consist of the members of $\mathcal{P}$ that change only finitely many coordinates and fix coordinates $1, \dots, N$. Then*

$$\sup_{S \in \mathcal{T}_N} h(\nu, S_{\#}\nu) \longrightarrow 0. \quad (51)$$

Also, the one-coordinate marginals of ν converge to the common coordinate law of μ , uniformly over coordinates greater than N .

Proof. Put $\ell = d\nu/d\mu$ and $\ell_N = \mathbb{E}_{\mu}[\ell \mid \sigma(x_1, \dots, x_N)]$. The martingale convergence theorem gives $\|\ell - \ell_N\|_1 \rightarrow 0$. Every $S \in \mathcal{T}_N$ preserves μ and satisfies $\ell_N \circ S^{-1} = \ell_N$. Thus

$$h(\nu, S_{\#}\nu)^2 \leq \|\ell - \ell \circ S^{-1}\|_{L^1(\mu)} \leq 2\|\ell - \ell_N\|_{L^1(\mu)},$$

which proves (51). For $i > N$, the marginal density is $\mathbb{E}_{\mu}[\ell \mid x_i]$, while independence gives $\mathbb{E}_{\mu}[\ell_N \mid x_i] = 1$, so

$$\|\mathbb{E}_{\mu}[\ell \mid x_i] - 1\|_{L^1(\mu)} \leq \|\ell - \ell_N\|_{L^1(\mu)}.$$

Taking the supremum over $i > N$ proves the marginal assertion. $\square$

Lemma 3.10 (Compactness of the covariance defect). *Let T be bounded on ℓ^2 . If $\nu_T \ll \mu$, then*

$$TT^* - I \text{ is compact.}$$

*If T is boundedly invertible, $T^*T - I$ is compact as well.*

Proof. We first show that, for every $B < \infty$, the squares of

$$Y_b := \sum_j b_j X_j, \quad \|b\|_2 \leq B, \quad (52)$$

are uniformly integrable. Let

$$X^{(M)} = X \mathbf{1}_{\{|X| \leq M\}} - \mathbb{E}[X \mathbf{1}_{\{|X| \leq M\}}], \quad Y_b^{(M)} = \sum_j b_j X_j^{(M)}.$$

The variables $X^{(M)}$ are bounded and centered, and converge to X in L^2 . Independence gives

$$\sup_{\|b\|_2 \leq B} \|Y_b - Y_b^{(M)}\|_2 \leq B \|X - X^{(M)}\|_2 \longrightarrow 0.$$

For fixed M , the fourth-moment formula for independent centered variables gives

$$\sup_{\|b\|_2 \leq B} \mathbb{E}|Y_b^{(M)}|^4 < \infty.$$

It also makes the finite partial sums Cauchy in L^4 , so the bound holds for countable sums. Moreover,

$$\sup_{\|b\|_2 \leq B} \|Y_b^2 - (Y_b^{(M)})^2\|_1 \leq B^2 \|X - X^{(M)}\|_2 (\|X\|_2 + \|X^{(M)}\|_2) \longrightarrow 0.$$

Thus Y_b^2 is uniformly approximated in L^1 by a uniformly integrable family, proving the claim using only the second moment of X .

Write $\nu = \nu_T$, $\ell = d\nu/d\mu$, and $\ell_N = \mathbb{E}_\mu[\ell \mid \sigma(x_1, \dots, x_N)]$. For a finitely supported unit vector a supported on coordinates greater than N , put $F_a(x) = (\sum_i a_i x_i)^2$. Under μ , these are squares from (52); under ν , they are the squares of Y_{T^*a} under μ . Since $\|T^*a\|_2 \leq \|T\|_{\text{op}}$, they are uniformly integrable under both laws. The bounded statistic $F_a \wedge L$ depends only on tail coordinates, so independence gives

$$|\mathbb{E}_\nu(F_a \wedge L) - \mathbb{E}_\mu(F_a \wedge L)| \leq L\|\ell - \ell_N\|_1.$$

Letting first $N \rightarrow \infty$ and then $L \rightarrow \infty$, uniformly in a , gives

$$\sup_{\substack{\|a\|_2=1, \text{ } a \text{ finitely supported} \\ \text{supp } a \subset \{N+1, N+2, \dots\}}} |\langle (TT^* - I)a, a \rangle| \rightarrow 0.$$

Since $TT^* - I$ is self-adjoint, density of finite vectors in the tail subspace implies

$$\|(I - P_N)(TT^* - I)(I - P_N)\|_{\text{op}} \rightarrow 0.$$

The complementary rows and columns are finite rank, so $TT^* - I$ is compact. If T is invertible, then $T^*T - I = T^{-1}(TT^* - I)T$. $\square$

Covariance compactness and unit-row rigidity match remote rows and columns with distinct allowed coordinates, but do not sum the errors. If their sum diverges, a remote packet can be swapped with better-matched auxiliary coordinates. After conjugation this gives $W = PT^{-1}ST$, whose Hilbert–Schmidt size is comparable to the packet error. Local coercivity detects W , whereas absolute continuity makes the remote output swap asymptotically invisible.

Lemma 3.11 (Hilbert–Schmidt necessity under a second-moment hypothesis). *Assume that the common law has an a.e.-positive density f , that $g = \sqrt{f} \in W^{1,2}(\mathbb{R})$, and that $xg' \in L^2(\mathbb{R})$. If $T \in \text{GL}(H)$ and $\nu_T \sim \mu$, then*

$$T - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}. \quad (53)$$

When T is orthogonal, the condition $xg' \in L^2$ can be omitted.

Proof. By lemma 3.10, $\|T^*e_i\|_2 \rightarrow 1$, and by lemma 3.9, the law of the corresponding row sum tends to ρ . Normalization changes that sum by $o_{L^2}(1)$, so lemma 3.8 gives indices $\iota(i)$ and ρ -preserving signs ς_i such that

$$\widehat{e}_i := \varsigma_i e_{\iota(i)}, \quad \|T^*e_i - \widehat{e}_i\|_2^2 \rightarrow 0. \quad (54)$$

The indices $\iota(i)$ are eventually distinct: if $\iota(i) = \iota(j)$, then $\varsigma_i T^*e_i - \varsigma_j T^*e_j$ has arbitrarily small norm by (54), whereas bounded invertibility gives the lower bound $\sqrt{2}/\|T^{-1}\|_{\text{op}}$. Fix N_0 so that ι is injective on $i > N_0$.

Since $\iota(i)$ leaves every finite set, compactness of $T^*T - I$ gives

$$\|T\widehat{e}_i\|_2^2 \rightarrow 1.$$

Together with $\langle T\widehat{e}_i, e_i \rangle = \langle \widehat{e}_i, T^*e_i \rangle \rightarrow 1$, this yields

$$\text{err}_i := \|T^*e_i - \widehat{e}_i\|_2^2 + \|T\widehat{e}_i - e_i\|_2^2 \rightarrow 0. \quad (55)$$

We claim that $\sum_{i>N_0} \text{err}_i < \infty$. Choose the local constants in [lemma 3.6](#) so that

$$h(\mu, \nu_W) \geq c_{\text{HS}} \|W - I\|_{\mathcal{S}_2} \quad \text{when } W \in \text{GL}(H), \quad \|W - I\|_{\mathcal{S}_2} \leq \delta_{\text{HS}}. \quad (56)$$

Suppose the claim fails, and choose $\delta > 0$ so that $3\|T^{-1}\|_{\text{op}}\delta < \delta_{\text{HS}}$. In an arbitrarily remote tail where $\text{err}_i \leq \delta^2$, choose a finite set E satisfying

$$\delta^2 \leq \sum_{i \in E} \text{err}_i \leq 2\delta^2. \quad (57)$$

Pair each $i \in E$ with a distinct later index k_i such that, for $G = \{k_i : i \in E\}$, the sum $\sum_{k \in G} \text{err}_k$ is arbitrarily small.

Let S exchange each output pair (e_i, e_{k_i}) , and let P exchange the aligned signed input pairs $(\hat{e}_i, \hat{e}_{k_i})$, acting identically on their orthogonal complements. Both are involutions in $\mathcal{P}$, so they preserve μ . Define

$$W = PT^{-1}ST.$$

Then

$$W - I = PT^{-1}(ST - TP), \quad \frac{\|ST - TP\|_{\mathcal{S}_2}}{\|T\|_{\text{op}}} \leq \|W - I\|_{\mathcal{S}_2} \leq \|T^{-1}\|_{\text{op}} \|ST - TP\|_{\mathcal{S}_2}. \quad (58)$$

Hellinger invariance and $P_{\#}\mu = \mu$ give

$$h(\mu, \nu_W) = h(\nu_T, S_{\#}\nu_T). \quad (59)$$

All differences have finite rank, and [lemma 3.1](#) supplies the common measurable carrier needed for this identity.

To compare the norms, freeze the auxiliary matches by defining

$$T_0^* e_k = \hat{e}_k, \quad T_0 \hat{e}_k = e_k \quad (k \in G).$$

On the complementary block let T_0 agree with T . Orthogonal block decomposition gives

$$\|T - T_0\|_{\mathcal{S}_2}^2 \leq \sum_{k \in G} (\|T^* e_k - \hat{e}_k\|_2^2 + \|T \hat{e}_k - e_k\|_2^2) = \sum_{k \in G} \text{err}_k. \quad (60)$$

Using the decompositions $E \oplus (E \cup G)^c \oplus G$ on the output and the corresponding aligned input coordinates, write

$$T_0 = \begin{pmatrix} A & B & 0 \\ C & D & 0 \\ 0 & 0 & I \end{pmatrix}, \quad ST_0 - T_0P = \begin{pmatrix} 0 & -B & I - A \\ C & 0 & -C \\ A - I & B & 0 \end{pmatrix}.$$

Here B, C are Hilbert–Schmidt, and the error on E is

$$\sum_{i \in E} (\|T_0^* e_i - \hat{e}_i\|_2^2 + \|T_0 \hat{e}_i - e_i\|_2^2) = 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2,$$

while the commutator satisfies

$$\|ST_0 - T_0P\|_{\mathcal{S}_2}^2 = 2(\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2),$$

Therefore,

$$\begin{aligned} 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2 &\leq \|ST_0 - T_0P\|_{\mathcal{S}_2}^2 \\ &\leq 2(2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2). \end{aligned} \quad (61)$$

As $\sum_{k \in G} \text{err}_k \rightarrow 0$, the first displayed error tends to $\sum_{i \in E} \text{err}_i$, while

$$\|(ST - TP) - (ST_0 - T_0P)\|_{\mathcal{S}_2} \leq 2\|T - T_0\|_{\mathcal{S}_2} \rightarrow 0.$$

Hence (57) and (61) allow the auxiliary indices to be chosen so that

$$\frac{\delta}{2} \leq \|ST - TP\|_{\mathcal{S}_2} \leq 3\delta.$$

It follows from (58) that

$$\frac{\delta}{2\|T\|_{\text{op}}} \leq \|W - I\|_{\mathcal{S}_2} \leq 3\|T^{-1}\|_{\text{op}}\delta < \delta_{\text{HS}}.$$

By (56), $h(\mu, \nu_W) \geq c_{\text{HS}}\delta/(2\|T\|_{\text{op}})$. But S is supported in the chosen remote tail, so lemma 3.9 and (59) make this quantity tend to zero as the tail recedes. This contradiction proves $\sum_{i > N_0} \text{err}_i < \infty$.

Define Q_0 by the matched rows $\hat{e}_i$ after the fixed initial cutoff N_0 , and by zero rows for $i \leq N_0$. The summability just proved implies

$$\begin{aligned} \|T - Q_0\|_{\mathcal{S}_2}^2 &= \sum_{i \leq N_0} \|T^*e_i\|_2^2 + \sum_{i > N_0} \|T^*e_i - \hat{e}_i\|_2^2 \\ &\leq \sum_{i \leq N_0} \|T^*e_i\|_2^2 + \sum_{i > N_0} \text{err}_i < \infty. \end{aligned}$$

Thus Q_0 is a compact perturbation of the invertible operator T , hence Fredholm of index zero. Its cokernel is the span of the first N_0 output coordinates and its kernel is spanned by the unused input coordinates, so exactly N_0 columns are unused. Matching them to the first N_0 rows with positive signs completes Q_0 to $Q \in \mathcal{P}$ and proves (53).

If T is orthogonal, both covariance defects vanish, and every $W = PT^*ST$ above is orthogonal. Thus only the orthogonal part of lemma 3.6 is used, and it requires no scale Fisher information. $\square$

Exact linear stabilizer. Exact invariance requires only the second moment.

Proof of corollary 1.2. Every member of $\mathcal{P}$ preserves the product law. Conversely, if $\nu_T = \mu$, covariance gives $TT^* = I$. Since T is boundedly invertible, it is orthogonal. Each of its unit rows gives a linear combination with the same law as X . Applying lemma 3.8 to the constant sequence consisting of this row shows that the row is an allowed signed unit vector. Orthogonality makes the selected coordinates a bijection. Thus $T \in \mathcal{P}$. $\square$

Proof of theorem 1.1. Necessity follows from lemma 3.11. If $T - Q \in \mathcal{S}_2$ for $Q \in \mathcal{P}$, then $TQ^{-1} = I + (T - Q)Q^{-1}$ is invertible and differs from the identity by a Hilbert–Schmidt operator. By proposition 3.4, $\nu_{TQ^{-1}} \sim \mu$. Since $Q_{\#}\mu = \mu$, this is $\nu_T \sim \mu$. For $K = TQ^{-1} - I$, the same proposition shows that the likelihood ratios for $I + P_n K P_n$ converge to $d\nu_T/d\mu$ in $L^1(\mu)$, and their square roots converge in $L^2(\mu)$. Lemmas 3.2 and 3.3 give the projected-energy characterization, the singular alternative, and two-sided uniform integrability. The exact linear stabilizer was proved above. $\square$

Proof of corollary 1.3. Necessity is lemma 3.11. For sufficiency, write $UQ^{-1} = I + (U - Q)Q^{-1}$. This is orthogonal and differs from the identity by a Hilbert–Schmidt operator. Apply proposition 3.5, and then use $Q_{\#}\mu = \mu$. The alternative follows from lemma 3.2. $\square$

3.6 Globalization of finite-dimensional coercivity

To globalize local coercivity, stack a hypothetical sequence of finite-dimensional counterexamples with summable Hellinger energies into one block-diagonal operator. Kakutani gives equivalence, so the classification makes this operator Hilbert–Schmidt-close to an allowed signed coordinate permutation; that permutation must eventually preserve the blocks, a contradiction.

Proof of theorem 2.7. Fix $0 < a \leq b < \infty$. We first prove uniform separation away from the exact linear stabilizer: for every $\varepsilon > 0$,

$$\inf \left\{ h(\mu_n, A_{\#}\mu_n)^2 : \begin{array}{l} n \geq 1, A \in \text{GL}(n, \mathbb{R}), \\ s_{\min}(A) \geq a, s_{\max}(A) \leq b, \\ d_n(A) \geq \varepsilon \end{array} \right\} > 0. \quad (62)$$

If this fails, choose matrices $A_k \in \text{GL}(n_k, \mathbb{R})$ in the displayed class whose Hellinger distances tend to zero. Since

$$-2 \log A(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) = -2 \log \left(1 - \frac{1}{2} h(\mu_{n_k}, (A_k)_{\#}\mu_{n_k})^2 \right),$$

we may pass to a subsequence for which the sum of these block energies is finite.

On consecutive coordinate blocks define $A = \bigoplus_{k \geq 1} A_k \in \text{GL}(H)$. Each block law is equivalent to μ_{n_k} , and the energy summability gives

$$\prod_{k \geq 1} A(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) > 0.$$

Kakutani’s theorem [17] gives $\nu_A \sim \mu$, so theorem 1.1 supplies $Q \in \mathcal{P}$ with

$$D := A - Q \in \mathcal{S}_2(H). \quad (63)$$

Let Π_k be the orthogonal projection onto the k -th output block. The orthogonality of these blocks gives

$$\sum_k \|\Pi_k D\|_{\mathcal{S}_2}^2 = \|D\|_{\mathcal{S}_2}^2 < \infty, \quad \text{hence} \quad \|\Pi_k D\|_{\mathcal{S}_2} \longrightarrow 0.$$

If e_i belongs to the k th output block but Q^*e_i lies outside the corresponding input block, then the two vectors have disjoint supports and

$$\|A^*e_i - Q^*e_i\|_2^2 = \|A_k^*e_i\|_2^2 + 1 \geq a^2 + 1. \quad (64)$$

For large k this contradicts $\|\Pi_k D\|_{\mathcal{S}_2}^2 < a^2 + 1$. Thus Q^* preserves every sufficiently large block; its restriction is a bijection and defines $Q_k \in \mathcal{P}_{n_k}$. Hence

$$d_{n_k}(A_k) \leq \|A_k - Q_k\|_F = \|\Pi_k D\|_{\mathcal{S}_2} \longrightarrow 0,$$

contrary to $d_{n_k}(A_k) \geq \varepsilon$. This proves (62).

Let $c_{\text{loc}}, C_{\text{loc}}, \delta_{\text{loc}}$ be the local constants, with $0 < \delta_{\text{loc}} < 1$. Choose a minimizer $Q_n \in \mathcal{P}_n$ for $d_n(T)$. Since $(Q_n)_{\#}\mu_n = \mu_n$,

$$h(\mu_n, T_{\#}\mu_n) = h(\mu_n, (TQ_n^{-1})_{\#}\mu_n), \quad \|TQ_n^{-1} - I\|_F = d_n(T).$$

For $d_n(T) \leq \delta_{\text{loc}}$, the local theorem gives both bounds. For $d_n(T) > \delta_{\text{loc}}$, (62) gives the lower bound and $h^2 \leq 2$ the upper bound. Taking the smaller lower constant and $C = \max\{C_{\text{loc}}^2, 2\delta_{\text{loc}}^{-2}\}$ proves (33). $\square$

4 Symmetric Stable Products

For symmetric stable products, both the coefficient domain and the mixing cost are α -dependent. Marques and Cambanis treat product-preserving affine maps [23]; for Cauchy products, coordinatewise location and scale changes are characterized in [20, 26]. Off-diagonal dependence instead leads to the column cost below. Fix $0 < \alpha < 2$, let ρ_α be the standard symmetric α -stable law, and put

$$\int_{\mathbb{R}} e^{itx} \rho_\alpha(dx) = e^{-|t|^\alpha}, \quad \mu_\alpha = \rho_\alpha^{\otimes \mathbb{N}}, \quad \mu_{\alpha,n} = \rho_\alpha^{\otimes n}. \quad (65)$$

For this symmetric law, $\mathcal{P}$ is the full signed-permutation group and $\mathcal{P}_n$ its n -dimensional version. Let $(X_j)_{j \geq 1}$ be independent with common law ρ_α , and write $\mathbf{X} = (X_j)_{j \geq 1}$. We use the symmetric unit-scale convention of [25]. The density f_α is smooth and positive and has finite location and scale Fisher information despite its infinite second moment [4]: the density and derivative asymptotics in [40] give $f_\alpha(x) \asymp (1 + |x|)^{-1-\alpha}$ and location score $O((1 + |x|)^{-1})$. For $\alpha \leq 1$, the normalization in (65) specifies a zero location parameter, not a finite mean.

A stable series $\sum_j a_j X_j$ converges almost surely exactly for $a \in \ell^\alpha$ [30]. For a matrix T with such rows, let $\Phi_{\alpha,T}$ be its row-series map and $\nu_{\alpha,T} = (\Phi_{\alpha,T})_{\#}\mu_\alpha$. The realization is *compatible* if some matrix S , also with ℓ^α rows, has measurable row-series maps satisfying

$$\Phi_{\alpha,S} \circ \Phi_{\alpha,T} = I, \quad \Phi_{\alpha,T} \circ \Phi_{\alpha,S} = I$$

on a common full-measure carrier for the corresponding orbit laws. Unlike lemma 3.1, this is an assumption, not a consequence of boundedness on ℓ^2 .

For a matrix K , put $F_\alpha(K) = \sum_j \|Ke_j\|_2^\alpha$. For $M = (m_{ij})$, define

$$\mathfrak{e}_\alpha(M) = \sum_i |m_{ii} - 1|^2 + F_\alpha(M - \text{diag } M). \quad (66)$$

Thus scales cost quadratically, whereas off-diagonal input columns cost their ℓ^2 -norms to the α -th power. For the finite output marginals, set

$$E_{\alpha,n}(T) = -2 \log A(\mu_{\alpha,n}, (\pi_n)_{\#}\nu_{\alpha,T}). \quad (67)$$

Theorem 4.1 (Complete criterion for symmetric stable coordinates). *Let $0 < \alpha < 2$ and suppose that T admits a compatible stable row-series realization. Then*

$$\begin{aligned} \nu_{\alpha,T} \sim \mu_\alpha &\iff T|_H \in \text{GL}(H) \text{ and } \mathfrak{e}_\alpha(TQ^{-1}) < \infty \text{ for some } Q \in \mathcal{P} \\ &\iff \sup_n E_{\alpha,n}(T) < \infty, \end{aligned}$$

(68)

If these conditions fail, then $\nu_{\alpha,T} \perp \mu_\alpha$ and $E_{\alpha,n}(T) \uparrow \infty$. For $T \in \text{GL}(H)$, the displayed summability condition itself supplies compatible stable row-series realizations of T and T^{-1} . The exact linear stabilizer in this class is $\mathcal{P}$.

Proof. The proof separates the equivalence–singularity dichotomy, a Gaussian-mixture upper bound, and Fourier/rare-event arguments for necessity.

Dichotomy, energy, and the coefficient domain. Put $g_\alpha = \sqrt{f_\alpha}$ and define

$$a_\alpha(u) = \int_{\mathbb{R}} g_\alpha(x) g_\alpha(x - u) dx.$$

Finite positive location Fisher information and decay of translations in L^2 give

$$1 - a_\alpha(u) \asymp_\alpha \min\{u^2, 1\}. \quad (69)$$

Kakutani’s theorem therefore identifies ℓ^2 as the translation space of μ_α . Under equivalence, conjugating translations through the compatible realization shows that T and its inverse matrix S preserve ℓ^2 ; the closed graph theorem gives $T|_H \in \text{GL}(H)$.

The inverse also makes every finite family of output rows linearly independent; a nonsingular input minor convolved with the remaining inputs gives a positive finite output marginal density. Using compatibility for composition, (69) for translation quasi-invariance, and these marginal densities, the proofs of [lemmas 3.2](#) and [3.3](#) adapt to μ_α . Together with the equivalence–singularity theorem of [\[27\]](#), they give

$$\nu_{\alpha,T} \sim \mu_\alpha \iff \sup_n E_{\alpha,n}(T) < \infty, \quad (70)$$

with singularity and divergent energy on the other branch.

Sufficiency.

$$\begin{aligned} F_\alpha(AK) &\leq \|A\|_{\text{op}}^\alpha F_\alpha(K), & F_\alpha(KD) &\leq \|D\|_{\text{op}}^\alpha F_\alpha(K), \\ \|K^* \gamma\|_\alpha^\alpha &\leq F_\alpha(K) \|\gamma\|_2^\alpha, \end{aligned} \quad (71)$$

for bounded A , bounded diagonal D , and $\gamma \in \ell^2$. In particular, finite column cost gives admissible stable rows.

The finite-dimensional estimate that replaces the finite-variance Hellinger path bound is

$$h^2(A \# \mu_{\alpha,n}, \mu_{\alpha,n}) \leq C_\alpha \min\{\mathfrak{e}_\alpha(A), 1\} \quad (72)$$

for every invertible real $n \times n$ matrix A , with a constant independent of n . Put $\beta = \alpha/2$ and use the positive-stable representation $X_i = \sqrt{2V_i}Z_i$, where $\mathbb{E}e^{-tV_i} = e^{-t^\beta}$ and the Z_i are independent standard Gaussians [\[30, 31\]](#). Conditioning on $V = \text{diag}(V_i)$, Hellinger data processing and the Gaussian affinity formula bound the left side of (72) by a constant times

$$\mathbb{E} \min\{\|V^{-1/2}(A - I)V^{1/2}\|_F^2, 1\}.$$

For $c_j^2 = \sum_{i \neq j} |a_{ij}|^2$, source column j contributes $W_j = V_j \sum_{i \neq j} |a_{ij}|^2 / V_i$. The positive-stable tail and truncated-moment bounds, together with $\mathbb{E}V_1^{-1} < \infty$, give

$$\mathbb{P}(c_j^2 V_j > 1) + \mathbb{E}[W_j; c_j^2 V_j \leq 1] \leq C_\alpha c_j^{2\beta} = C_\alpha c_j^\alpha.$$

Summation over j , followed by the quadratic diagonal-scale estimate, proves (72).

Remove the measure-preserving Q and write $M = D + K$, where $D = \text{diag } M$ has entries d_i , $(d_i - 1) \in \ell^2$, and $F_\alpha(K) < \infty$. Only finitely many d_i are nonpositive. Replace each of them by 1 in D and add $d_i - 1$ to the corresponding diagonal entry of K ; this preserves both summability conditions and makes D positive and boundedly invertible. Set

$$D_n = I + P_n(D - I)P_n, \quad K_n = P_n K P_n, \quad M_n = D_n + K_n,$$

so that $M_n \rightarrow M$ in operator norm and the M_n are eventually invertible with uniformly bounded inverses. The identity $M_n^{-1} = D_n^{-1} - M_n^{-1} K_n D_n^{-1}$ and (71) give these inverses uniformly bounded diagonal parts and column costs. Applied to $M_n^{-1}(M_m - M_n)$, the same bounds show that its off-diagonal column cost and diagonal square cost tend to zero; here $F_\alpha(K_m - K_n) \rightarrow 0$, $\|D_m - D_n\|_{\text{op}} \rightarrow 0$, and finite column cost implies Hilbert–Schmidt membership. Hence (72) makes the square-root likelihoods of ν_{α, M_n} Cauchy in L^2 . The limit has norm one, and ℓ^α -convergence of every row identifies its square with $d\nu_{\alpha, M}/d\mu_\alpha$. This proves absolute continuity, which (70) upgrades to equivalence. The identity $(D + K)^{-1} = D^{-1} - (D + K)^{-1} K D^{-1}$ gives the same admissible form for the inverse. Continuity of stable random functionals from ℓ^α to convergence in probability then passes finite-matrix composition to the row-series limits and supplies compatible versions.

Reduction to a Hilbert–Schmidt perturbation. For necessity, suppose $\nu_{\alpha, T} \sim \mu_\alpha$. We first transfer the half-density through Fourier transform to obtain the Hilbert–Schmidt part; the sharper column cost will be recovered separately.

Let $\widehat{g}_\alpha$ be the unitary Fourier transform of g_α , and set

$$\rho_\alpha^\sharp(d\xi) = |\widehat{g}_\alpha(\xi)|^2 d\xi.$$

The law $\rho_\alpha^\sharp$ is symmetric, has variance $\|g'_\alpha\|_2^2 \in (0, \infty)$, and has characteristic function a_α . Since $\widehat{g}_\alpha * \widehat{g}_\alpha$ is a positive multiple of $e^{-|t|^\alpha}$, we also have $|\widehat{g}_\alpha|^2 * |\widehat{g}_\alpha|^2 > 0$ almost everywhere. Polynomial stable tails give two-sided polynomial bounds for a_α , and a'_α is bounded. Choose an integer r so large that

$$\int_{\mathbb{R}} t^2 (|a_\alpha(t)|^{2r} + |(a'_\alpha)^r(t)|^2) dt < \infty.$$

Fourier inversion gives $(\rho_\alpha^\sharp)^{*r}$ a bounded-variation density, whose third convolution has finite location Fisher information by the smoothing estimate in [4]. If f_{3r} is the density of $(\rho_\alpha^\sharp)^{*3r}$, U_1, U_2 are independent with this density, and s_{3r}, s_{6r} are the scores of f_{3r} and $f_{3r} * f_{3r}$, then

$$xs_{6r}(x) - 1 = 2\mathbb{E}[U_1 s_{3r}(U_2) \mid U_1 + U_2 = x].$$

Thus finite variance and location Fisher information give finite scale Fisher information after one further convolution. Set $k = 6r$ and $\mu_{\alpha, k}^\sharp = ((\rho_\alpha^\sharp)^{*k})^{\otimes \mathbb{N}}$. After rescaling to variance one, $(\rho_\alpha^\sharp)^{*k}$ has an a.e.-positive density satisfying (1); it is non-Gaussian because its characteristic function a_α^k has a polynomial lower bound.

Let $f_{\alpha, T, n}$ be the density of the first n output marginal and put

$$\ell_{\alpha, T, n} = \frac{f_{\alpha, T, n}}{f_\alpha^{\otimes n}}, \quad \ell_{\alpha, T} = \frac{d\nu_{\alpha, T}}{d\mu_\alpha}.$$

Here $\ell_{\alpha, T, n}$ is lifted to the product space and $\sqrt{f_{\alpha, T, n}} = g_\alpha^{\otimes n} \sqrt{\ell_{\alpha, T, n}}$. Equivalence gives $\sqrt{\ell_{\alpha, T, n}} \rightarrow \sqrt{\ell_{\alpha, T}}$ in $L^2(\mu_\alpha)$.

Fourier transformation turns half-density translations into phases; tensorization and frequency addition then produce the desired convolution. Precisely, take k tensor copies of $\sqrt{f_{\alpha,T,n}}$, Fourier transform, square the modulus, and add the k frequencies at each site. Let $\Xi_{\alpha,T,n}$ be the product of the resulting law on $\mathbb{R}^n$ with $((\rho_\alpha^\#)^{*k})^{\otimes \{i>n\}}$. The first factor has a Lebesgue density and the tail density is positive almost everywhere; hence $\Xi_{\alpha,T,n} \ll \mu_{\alpha,k}^\#$.

For $m > n$, pad $\sqrt{f_{\alpha,T,n}}$ by $m - n$ factors of g_α . Unitarity, contraction under pushforward, and $\| |u|^2 - |v|^2 \|_1 \leq 2\|u - v\|_2$ give

$$\|\Xi_{\alpha,T,m} - \Xi_{\alpha,T,n}\|_{\text{TV}} \leq 2\|(\sqrt{\ell_{\alpha,T,m}})^{\otimes k} - (\sqrt{\ell_{\alpha,T,n}})^{\otimes k}\|_{L^2(\mu_\alpha^{\otimes k})} \leq 2k\|\sqrt{\ell_{\alpha,T,m}} - \sqrt{\ell_{\alpha,T,n}}\|_{L^2(\mu_\alpha)}.$$

Hence $\Xi_{\alpha,T,n}$ converges in total variation to a probability $\Xi_{\alpha,T} \ll \mu_{\alpha,k}^\#$.

Let $\nu_{\alpha,T}^{[n]} = \ell_{\alpha,T,n} \mu_\alpha$. For every finitely supported z , Fourier autocorrelation and square-root convergence give

$$\begin{aligned} \widehat{\Xi}_{\alpha,T}(z) &= \lim_n \mathbf{A}(\nu_{\alpha,T}^{[n]}, (x \mapsto x + z)_\# \nu_{\alpha,T}^{[n]})^k \\ &= \mathbf{A}(\nu_{\alpha,T}, (x \mapsto x + z)_\# \nu_{\alpha,T})^k \\ &= \prod_i a_\alpha((T^{-1}z)_i)^k. \end{aligned}$$

The last expression is the characteristic function of $(\Phi_{(T^{-1})^*})_\# \mu_{\alpha,k}^\#$; agreement for finitely supported z identifies all finite marginals. Hence

$$(\Phi_{(T^{-1})^*})_\# \mu_{\alpha,k}^\# \ll \mu_{\alpha,k}^\#. \quad (73)$$

The dichotomy and classification in [theorem 1.1](#) upgrade this to equivalence and give $Q \in \mathcal{P}$ such that $(T^{-1})^* - Q \in \mathcal{S}_2$. Since

$$T - Q = -T((T^{-1})^* - Q)^*Q,$$

we obtain $T - Q \in \mathcal{S}_2$. After removing Q , write

$$TQ^{-1} = I + K, \quad K \in \mathcal{S}_2. \quad (74)$$

Necessity of the off-diagonal column summability. For $\alpha < 2$, Hilbert–Schmidt control is weaker than the column cost. We detect column j on the event $|X_j| \asymp \|Be_j\|_2^{-1}$, whose probability has the required order. We prove that there are constants $\varepsilon_\alpha, c_\alpha > 0$ such that, for every n and every $B \in \mathbb{R}^{n \times n}$,

$$\text{diag } B = 0, \quad F_\alpha(B) \leq \varepsilon_\alpha \implies h^2((I + B)_\# \mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha F_\alpha(B). \quad (75)$$

To prove it, write $F = F_\alpha(B)$, $c_j = \|Be_j\|_2$, and $u_j = Be_j/c_j$ when $c_j \neq 0$. Then $\|u_j\|_2 = 1$, $(u_j)_j = 0$, and the stable scale contributed to output row i satisfies

$$\omega_i(B) := \sum_j |b_{ij}|^\alpha \leq F. \quad (76)$$

For a finite family of Hilbert-space vectors v_j , excluding the events $\{\|v_j\|_2 |X_j| > t\}$ and applying Chebyshev to the centered truncated sum gives a tail of order $t^{-\alpha} \sum_j \|v_j\|_2^\alpha$. Integration yields a modulus $\tau_\alpha(s) \downarrow 0$ such that

$$\mathbb{E} \min \left\{ \left\| \sum_j v_j X_j \right\|_2, 1 \right\} \leq \tau_\alpha \left(\sum_j \|v_j\|_2^\alpha \right). \quad (77)$$

Next, a bounded statistic detects unit displacements uniformly in the dimension. Set $\psi(x) = \tanh x$. There are a smooth odd increasing χ , bounded by one, and $c_\alpha > 0$ such that every unit vector u satisfies

$$\Theta_u(x) = \sum_i u_i \psi(x_i), \quad \mathcal{M}_u(t) = \mathbb{E} \chi(\Theta_u(\mathbf{X} + tu)), \quad \mathcal{M}_u(t) \geq c_\alpha \quad (1 \leq t \leq 2). \quad (78)$$

Indeed, $\Theta_u(\mathbf{X})$ has mean zero and variance $\text{Var}(\psi(X))$, while its shift by tu changes it by at most 2 for $|t| \leq 2$. A fixed compact interval therefore carries all but uniformly small mass. Choose χ' positive on that interval enlarged by two. The bound $\inf_{|\zeta| \leq 2} \mathbb{E} \psi'(X + \zeta) > 0$ then makes $\mathcal{M}'_u(t)$ uniformly positive after the discarded contribution is absorbed. Oddness gives $\mathcal{M}_u(0) = 0$, proving (78).

Now choose a nonzero smooth odd function w , supported where $1 \leq |t| \leq 2$, with $0 \leq w \leq 1$ on the positive half-line. Put

$$\Theta_j(x) = \sum_{i \neq j} (u_j)_i \psi(x_i),$$

and define the bounded test

$$\Psi_B(x) = \max \left\{ -1, \min \left\{ \sum_{j: c_j \neq 0} w(c_j x_j) \chi(\Theta_j(x)), 1 \right\} \right\}.$$

For all sufficiently small c_j , stable tail asymptotics give

$$\mathbb{P}(1 \leq |c_j X_j| \leq 2) \leq C_\alpha c_j^\alpha, \quad \mathbb{E}|w(c_j X_j)| \geq c_\alpha c_j^\alpha. \quad (79)$$

The events $\{w(c_j X_j) \neq 0\}$ are independent, and the probability that at least two occur is $O_\alpha(F^2)$. Before truncation each summand has mean zero because $w(c_j X_j)$ is independent of $\Theta_j(\mathbf{X})$; hence $|\mathbb{E} \Psi_B(\mathbf{X})| \leq C_\alpha F^2$.

Set $\mathbf{Y} = (I + B)\mathbf{X}$ and fix $0 < \delta < 1/4$. By (76),

$$\mathbb{P}\{\exists i : |c_i(B\mathbf{X})_i| > \delta\} \leq C_\alpha \delta^{-\alpha} \sum_i c_i^\alpha \omega_i(B) \leq C_\alpha \delta^{-\alpha} F^2.$$

On the complementary event, Lipschitz continuity of w and (79) show that replacing every factor $w(c_j Y_j)$ by $w(c_j X_j)$ changes the expectation by at most $C_\alpha(\delta F + \delta^{-\alpha} F^2)$; removing the outer truncation costs $O_\alpha(F^2)$. The remaining-coordinate perturbation $B(I - e_j \otimes e_j)\mathbf{X}$ is independent of X_j , and (77) shows that replacing $\Theta_j(\mathbf{Y})$ by $\Theta_j(\mathbf{X} + c_j X_j u_j)$ costs at most $C_\alpha F \tau_\alpha(F)$ after summation. Because $(u_j)_j = 0$, conditioning on X_j and using Equations (78) and (79) gives

$$\sum_j \mathbb{E}[w(c_j X_j) \chi(\Theta_j(\mathbf{X} + c_j X_j u_j))] \geq c_\alpha F.$$

Thus

$$\mathbb{E} \Psi_B(\mathbf{Y}) - \mathbb{E} \Psi_B(\mathbf{X}) \geq c_\alpha F - C_\alpha \{\delta F + (1 + \delta^{-\alpha}) F^2 + F \tau_\alpha(F)\}.$$

Choose δ , then F , so that the error is at most half the main response. The test is nonzero only when at least one such event occurs, and the coordinates of $\mathbf{Y}$ have stable scales at most $(1 + F)^{1/\alpha}$; hence

$$\mathbb{E} \Psi_B(\mathbf{X})^2 + \mathbb{E} \Psi_B(\mathbf{Y})^2 \leq C_\alpha F.$$

Hellinger Cauchy–Schwarz now gives (75).

We globalize this local estimate using finite tail sign changes. Such a change isolates the entries crossing a coordinate cut: equivalence makes every remote cut Hellinger-invisible, whereas the local estimate detects every cut of a fixed small cost.

Let $B \in \mathcal{S}_2$ have ℓ^α rows, zero diagonal, and $\|B\|_{\text{op}} \leq 1/10$. Suppose $I + B$ admits a compatible stable row-series realization and $\nu_{\alpha, I+B} \sim \mu_\alpha$. For a finite coordinate set $\mathcal{J}$, let $Q_{\mathcal{J}}$ change the signs in $\mathcal{J}$ and set

$$\Delta_{\mathcal{J}} = Q_{\mathcal{J}} B Q_{\mathcal{J}} - B, \quad A_{\mathcal{J}} = (I + B)^{-1} Q_{\mathcal{J}} (I + B) Q_{\mathcal{J}}.$$

The matrix $\Delta_{\mathcal{J}}$ consists of the entries crossing the cut. Since $\mathcal{J}$ is finite, $\Delta_{\mathcal{J}}$ has finite column cost. Applying (75) to $P_n \Delta_{\mathcal{J}} P_n$ and using the sufficiency approximation extends that bound to $\Delta_{\mathcal{J}}$.

Put

$$K_{\mathcal{J}} = (I + B)^{-1} \Delta_{\mathcal{J}}, \quad D_{\mathcal{J}} = I + \text{diag } K_{\mathcal{J}}, \quad \tilde{B}_{\mathcal{J}} = D_{\mathcal{J}}^{-1} (I + K_{\mathcal{J}}) - I.$$

Then $A_{\mathcal{J}} = I + K_{\mathcal{J}}$ and $\text{diag } \tilde{B}_{\mathcal{J}} = 0$. Since $\|(I + B)^{-1} - I\|_{\text{op}} \leq 1/9$, projection away from each diagonal entry compares the off-diagonal columns of $K_{\mathcal{J}}$ and $\Delta_{\mathcal{J}}$. For small $F_\alpha(\Delta_{\mathcal{J}})$, summing these columnwise comparisons gives

$$F_\alpha(\tilde{B}_{\mathcal{J}}) \asymp_\alpha F_\alpha(\Delta_{\mathcal{J}}), \quad \|\text{diag } K_{\mathcal{J}}\|_2 \leq \frac{1}{9} F_\alpha(\Delta_{\mathcal{J}})^{1/\alpha}.$$

The lower bound for $\tilde{B}_{\mathcal{J}}$, the diagonal scale upper bound for $D_{\mathcal{J}}$, and the Hellinger triangle inequality now give

$$F_\alpha(\Delta_{\mathcal{J}}) \leq \varepsilon_* \implies h(\nu_{\alpha, A_{\mathcal{J}}}, \mu_\alpha) \geq c_* F_\alpha(\Delta_{\mathcal{J}})^{1/2}. \quad (80)$$

Indeed, by (75), the two terms have respective orders $F_\alpha(\Delta_{\mathcal{J}})^{1/2}$ and $F_\alpha(\Delta_{\mathcal{J}})^{1/\alpha}$, and the latter is smaller for small cost because $\alpha < 2$.

The finite-prefix argument in lemma 3.9, applied to the square-root likelihood, is uniform over finite tail sign changes. Since each such change preserves μ_α , Hellinger invariance gives

$$\sup_{\substack{\mathcal{J} \subset \{N+1, N+2, \dots\} \\ |\mathcal{J}| < \infty}} h(\mu_\alpha, \nu_{\alpha, A_{\mathcal{J}}}) \longrightarrow 0. \quad (81)$$

Suppose $F_\alpha(B) = \infty$. Deleting finitely many columns or rows changes the cost finitely, the latter because each row lies in ℓ^α ; thus every tail has infinite cost. For a finite tail set $\mathcal{I}_0$, include each coordinate independently in $\mathcal{J}$ with probability one half. Columnwise concavity of $t^{\alpha/2}$ gives

$$\mathbb{E}_{\mathcal{J}} F_\alpha(P_{\mathcal{I}_0} \Delta_{\mathcal{J}} P_{\mathcal{I}_0}) \geq 2^{\alpha-1} F_\alpha(P_{\mathcal{I}_0} B P_{\mathcal{I}_0}). \quad (82)$$

As $\mathcal{I}_0$ exhausts the tail, some finite cuts have arbitrarily large cost. Equivalence makes the remote one-coordinate output marginals converge to ρ_α , so $\omega_i(B) \rightarrow 0$ in the notation of (76); Hilbert–Schmidt membership also gives $c_i \rightarrow 0$. Hence

$$F_\alpha(\Delta_{\{i\}}) = 2^\alpha (c_i^\alpha + \omega_i(B)) \longrightarrow 0.$$

For $\alpha < 1$, $F_\alpha^{1/\alpha}$ is a quasi-norm and its α -th power F_α is subadditive; for $\alpha \geq 1$, $F_\alpha^{1/\alpha}$ is a norm. Since adding i to a cut changes $\Delta_{\mathcal{J}}$ by a sign conjugate of $\Delta_{\{i\}}$, the increments of F_α in the

first case and of $F_\alpha^{1/\alpha}$ in the second tend uniformly to zero. Fix $0 < \varepsilon < \varepsilon_*/2$, add the elements of a large cut from (82), and stop at the first threshold crossing. In every sufficiently remote tail this produces $\mathcal{J}$ with $\varepsilon \leq F_\alpha(\Delta_{\mathcal{J}}) \leq 2\varepsilon$. Equations (80) and (81) then contradict one another. Hence $F_\alpha(B) < \infty$.

Finally, for $M = I + K$ from (74), take $M_n = I + P_n K P_n$ and set $R = M_n^{-1} M$, which is close to I . Composition with the finite-coordinate map M_n^{-1} gives $\nu_{\alpha, R} \sim \mu_\alpha$. The diagonal part $D_R = \text{diag } R$ is positive, boundedly invertible, and satisfies $D_R - I \in \mathcal{S}_2$, so the one-coordinate scale criterion gives $\nu_{\alpha, D_R} \sim \mu_\alpha$. Thus

$$D_R^{-1} R = I + B, \quad \text{diag } B = 0, \quad B \in \mathcal{S}_2, \quad \|B\|_{\text{op}} \leq 1/10, \quad \nu_{\alpha, I+B} \sim \mu_\alpha.$$

Finite-coordinate and bounded diagonal composition preserve ℓ^α rows and compatibility; hence the finite-cut argument gives $F_\alpha(B) < \infty$. Since K_n is finitely supported,

$$M = M_n D_R (I + B) = D_R + (K_n D_R + D_R B + K_n D_R B).$$

The parenthesized term has finite column cost by (71). Restoring the signed permutation gives

$$\sum_j \left(\sum_{i \neq j} |k_{ij}|^2 \right)^{\alpha/2} < \infty,$$

while $K \in \mathcal{S}_2$ gives $\sum_i |k_{ii}|^2 < \infty$. Together these are $\mathfrak{e}_\alpha(TQ^{-1}) < \infty$.

If $\nu_{\alpha, T} = \mu_\alpha$, comparison of the joint characteristic functions, first for T and then for its compatible inverse, makes T^* a surjective isometry of ℓ^α . For $0 < \alpha < 2$, equality in

$$2\|u\|_\alpha^\alpha + 2\|v\|_\alpha^\alpha - \|u + v\|_\alpha^\alpha - \|u - v\|_\alpha^\alpha \geq 0$$

characterizes disjoint support. Hence the images of the coordinate vectors have disjoint supports; surjectivity makes each support a singleton, and norm preservation makes its entry ± 1 . Thus $T \in \mathcal{P}$. The converse follows from independence and symmetry. $\square$

4.1 Anisotropic stable Hellinger coercivity and sharpness

The two summands in (66) have genuinely different homogeneities. For a finite matrix A , put

$$\mathfrak{e}_{\alpha, n}(A) = \min_{Q \in \mathcal{P}_n} \mathfrak{e}_\alpha(AQ^{-1}).$$

Proposition 4.2 (Anisotropic stable Hellinger coercivity). *Let $A \in \text{GL}(n, \mathbb{R})$. If all singular values of A lie in a fixed interval $[a, b] \subset (0, \infty)$, then there are positive constants depending only on α, a, b , and not on the dimension, such that*

$$c_{\alpha, a, b} \min\{\mathfrak{e}_{\alpha, n}(A), 1\} \leq h^2(A \# \mu_{\alpha, n}, \mu_{\alpha, n}) \leq C_{\alpha, a, b} \min\{\mathfrak{e}_{\alpha, n}(A), 1\}. \quad (83)$$

Proof. The upper bound is (72), after removing the minimizing signed permutation. For the even convolution power k above, Fourier transformation and modulus contraction, followed by tensorization and pushforward under coordinatewise frequency addition, give

$$h^2(((A^{-1})^*)_{\#}((\rho_\alpha^\#)^{*k})^{\otimes n}, ((\rho_\alpha^\#)^{*k})^{\otimes n}) \leq k h^2(A \# \mu_{\alpha, n}, \mu_{\alpha, n}). \quad (84)$$

Apply finite-variance coercivity to the rescaled $(\rho_\alpha^\#)^{*k}$ product. The singular values of $(A^{-1})^*$ lie in $[b^{-1}, a^{-1}]$, and

$$A - Q = -A((A^{-1})^* - Q)^*Q$$

compares the two distances up to a factor depending on a, b . Hence

$$h^2(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_{\alpha,a,b} \min \left\{ \min_{Q \in \mathcal{P}_n} \|A - Q\|_F^2, 1 \right\}. \quad (85)$$

Near a signed permutation, remove that permutation and write

$$d_{\text{diag}} = \sum_i |a_{ii} - 1|^2, \quad F = \sum_j \left(\sum_{i \neq j} |a_{ij}|^2 \right)^{\alpha/2}.$$

For $d_{\text{diag}} + F$ small, the identity is the nearest signed permutation, so (85) controls d_{diag} . With $D = \text{diag } A$ and $D^{-1}A = I + B$, $\text{diag } B = 0$ and $F_\alpha(B) \asymp_\alpha F$. The off-diagonal column lower bound and the diagonal scale upper bound give

$$h(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha \sqrt{F} - C_\alpha \sqrt{d_{\text{diag}}}.$$

Choose $\eta \leq (c_\alpha/(2C_\alpha))^2$. Then $d_{\text{diag}} \leq \eta F$ implies $C_\alpha \sqrt{d_{\text{diag}}} \leq (c_\alpha/2)\sqrt{F}$. When $d_{\text{diag}} \geq \eta F$, use (85). This proves the local lower bound.

If the global lower bound failed, there would be $\varepsilon > 0$, dimensions n_k , and matrices $A_k \in \text{GL}(n_k, \mathbb{R})$, with singular values in $[a, b]$, such that $\mathfrak{e}_{\alpha, n_k}(A_k) \geq \varepsilon$ and, after passing to a subsequence,

$$\sum_k h^2((A_k)_{\#}\mu_{\alpha, n_k}, \mu_{\alpha, n_k}) < \infty.$$

The block-diagonal sum is boundedly invertible; its row map and inverse act blockwise and are compatible. Kakutani makes its image product equivalent to μ_α , so the classification supplies a signed permutation with finite anisotropic cost. A permutation column crossing a block contributes at least a^α , so all sufficiently late blocks are preserved. The corresponding block restrictions then give $\sum_k \mathfrak{e}_{\alpha, n_k}(A_k) < \infty$, contradicting the lower bound ε . This proves the global lower bound. $\square$

The different local powers already appear for the shear $T_t(x, y) = (x + ty, y)$. Conditioning on y and applying Plancherel gives

$$\begin{aligned} h^2((T_t)_{\#}\mu_{\alpha,2}, \mu_{\alpha,2}) &= 2 \int_{\mathbb{R}} (1 - e^{-|t\xi|^\alpha}) |\widehat{g}_\alpha(\xi)|^2 d\xi \\ &\sim 2|t|^\alpha \int_{\mathbb{R}} |\xi|^\alpha |\widehat{g}_\alpha(\xi)|^2 d\xi, \quad t \rightarrow 0. \end{aligned} \quad (86)$$

Diagonal scales instead cost t^2 , so no single finite-dimensional matrix Fisher quadratic form generates this anisotropic coercivity.

In particular, Hilbert–Schmidt proximity is necessary but no longer sufficient. On independent two-coordinate blocks, let

$$T_a(x_k, y_k)_{k \geq 1} = (x_k + a_k y_k, y_k)_{k \geq 1}.$$

Then $T_a - I \in \mathcal{S}_2$ exactly when $a \in \ell^2$, while [theorem 4.1](#) gives $\nu_{\alpha, T_a} \sim \mu_\alpha$ exactly when $a \in \ell^\alpha$. Any $a \in \ell^2 \setminus \ell^\alpha$ therefore gives a Hilbert–Schmidt perturbation with singular image law, even though the stable coordinate density is everywhere positive and has finite location and scale Fisher information. By contrast, when one source is mixed into many outputs, the criterion charges the ℓ^2 -norm of that whole column only once. An entrywise ℓ^α condition would therefore be too strong. The correct stable criterion is the anisotropic mixing cost [\(66\)](#), not a Schatten-class replacement for the finite-variance theorem.

5 Density Zeros and Support Rigidity

When the density has zeros, Hilbert–Schmidt proximity controls Hellinger accumulation, but equivalence also requires the forward and inverse zero-set row conditions: every row series of T and T^{-1} lies almost surely in $\{f > 0\}$. Bounded or one-sided support then forces permutation rigidity.

5.1 Equivalence criterion when the density has zeros

The Fourier/Fisher mechanism still supplies the metric part of the classification; for its matrix estimates the appropriate regularity is

$$\begin{aligned} f \geq 0, \quad \int f = 1, \quad g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad f \text{ non-Gaussian.} \end{aligned} \tag{87}$$

The global zero extension is essential: $g \in W^{1,2}(\mathbb{R})$ excludes boundary jumps and makes $g' = 0$ almost everywhere on $\{g = 0\}$. Thus $sg = -2g'$ and the Fourier/Fisher integrations by parts hold on the whole line without a boundary term. Interior regularity is insufficient; the uniform density has zero interior derivative but a jump in its zero-extended half-density.

Proposition 5.1 (Local Fourier coercivity with zeros of the density). *Under [\(87\)](#), the matrix Fisher operator decomposition, finite-dimensional Hellinger path bound, and dimension-free local Frobenius coercivity in [Section 2](#) remain valid. Write $\mu = \rho^{\otimes \mathbb{N}}$, $\rho(dx) = f(x)dx$. There are constants $c, C, \delta > 0$, with $\delta < 1$, such that*

$$c\|K\|_{\mathcal{S}_2} \leq h(\mu, \nu_{I+K}) \leq C\|K\|_{\mathcal{S}_2} \quad (\|K\|_{\mathcal{S}_2} \leq \delta). \tag{88}$$

Moreover, for every $T \in \text{GL}(H)$, the following necessary condition survives:

$$\nu_T \ll \mu \implies T - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}. \tag{89}$$

For orthogonal transformations one may omit $xg' \in L^2$, retaining non-Gaussianity, $g \in W^{1,2}(\mathbb{R})$, and normalization to a centered law with variance one. These statements do not assert absolute continuity of ν_{I+K} .

Proof. Define $s = -2g'/g$ on $\{g > 0\}$ and zero elsewhere. Since Sobolev derivatives vanish almost everywhere on zero sets, $sg = -2g'$ globally, and all cutoff identities—including $\mathbb{E}s(X) = 0$, $\mathbb{E}[Xs(X)] = 1$, and the formulas for J, κ —remain valid. The cases $J = 1$ and $\kappa = 0$ would respectively give $g' = -xg/2$ and $2xg' + g = 0$; the latter has no nonzero square-integrable solution. Thus the finite-dimensional proofs apply.

For $K \in \mathcal{S}_2$ with $I + K$ invertible, put $T_n = I + P_n K P_n$. The Hellinger path bound and invariance give, for large m, n ,

$$h(\nu_{T_n}, \nu_{T_m}) \leq C \|T_n^{-1}\|_{\text{op}} \|T_n - T_m\|_{\mathcal{S}_2}.$$

The square-root densities are therefore Cauchy in L^2 relative to a common dominating measure. Their norm-one limit is identified with ν_{I+K} by finite-row convergence, proving (88) but not $\nu_{I+K} \ll \mu$. Polar approximants and the skew-generator bound from [proposition 3.5](#) give the orthogonal case.

In [lemma 3.11](#), the tail-symmetry and covariance lemmas already require only $\nu_T \ll \mu$; the other inputs are moment-only rigidity, the measurable inverse action, and local coercivity for finite-rank tail perturbations. The estimate above supplies the last input, so the same tail-matching proof yields (89). $\square$

The zero-extended square root lies in $W^{1,2}(\mathbb{R})$ exactly for Gamma shape > 2 or both Beta shapes > 2 ; at shape one it jumps. The support results need no such regularity.

Theorem 5.2 (The linear equivalence group with density zeros). *Assume (87), let $\rho(dx) = f(x)dx$, and put $\mathcal{U} = \{x \in \mathbb{R} : f(x) > 0\}$. For every $T \in \text{GL}(H)$, the almost-sure requirements below are called the forward and inverse zero-set row conditions.*

$$\nu_T \ll \mu \iff \begin{cases} T - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}, \\ \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i. \end{cases} \quad (90)$$

Consequently,

$$\{T \in \text{GL}(H) : \nu_T \sim \mu\} = \left\{ T \in \text{GL}(H) : \begin{cases} T - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}, \\ \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0, \\ \mu \left\{ x : \sum_j (T^{-1})_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i \end{cases} \right\}.$$

It is a subgroup of $\text{GL}(H)$. For orthogonal members the criterion holds with $T^{-1} = T^*$ and without $xg' \in L^2$.

Whenever $T - Q \in \mathcal{S}_2(H)$ for some $Q \in \mathcal{P}$, the Lebesgue decomposition of ν_T relative to μ is its restriction to $\mathcal{U}^{\mathbb{N}}$ plus its restriction to the complement: the former is absolutely continuous and the latter is singular.

Proof. Suppose $T - Q \in \mathcal{S}_2$ and put $K = TQ^{-1} - I$. Since $\nu_Q = \mu$, $\nu_T = \nu_{I+K}$. For $T_n = I + P_n K P_n$, [proposition 5.1](#) gives $h(\nu_{T_n}, \nu_T) \rightarrow 0$, hence total-variation convergence. Each ν_{T_n} has a Lebesgue density in its first n coordinates and agrees with μ on the tail, so positivity of μ_n on $\mathcal{U}^n$ and Fubini give $\nu_{T_n}|_{\mathcal{U}^n} \ll \mu$. Passing to the limit proves the same for ν_T , while $\nu_T|_{(\mathcal{U}^{\mathbb{N}})^c} \perp \mu$. Consequently

$$\nu_T \ll \mu \iff \nu_T(\mathcal{U}^{\mathbb{N}}) = 1 \iff \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i.$$

Together with (89), this proves (90).

If $\nu_T \sim \mu$, the measurable inverse action gives $\nu_{T^{-1}} \sim \mu$, so the forward and inverse zero-set row conditions are necessary. Conversely, $T^{-1} - Q^{-1} = -T^{-1}(T - Q)Q^{-1} \in \mathcal{S}_2$, and these conditions with (90) give $\nu_T, \nu_{T^{-1}} \ll \mu$. The inverse action then yields

$$\mu = (\Phi_T)_\# \nu_{T^{-1}} \ll (\Phi_T)_\# \mu = \nu_T,$$

and hence equivalence. Measurable composition gives $\nu_{ST} = (\Phi_S)_\# \nu_T \sim \nu_S \sim \mu$ and closure under inverses follows above, proving the group assertion. In the orthogonal case, use the approximants from [proposition 3.5](#), which require only location Fisher information. $\square$

Null-set modifications of f do not affect the forward and inverse zero-set row conditions, since nonzero row sums have densities. They are automatic for positive densities; the rank-one contraction below satisfies the forward zero-set row condition but not the inverse zero-set row condition, yielding $\nu_T \ll \mu$ while $\mu \not\ll \nu_T$.

5.2 Boundary behavior of the density and summability criteria

Let ρ be uniform on $[-\sqrt{3}, \sqrt{3}]$, let $\mu_{u,2} = \rho^{\otimes 2}$, and consider $T = \bigoplus_k O_{\theta_k}$. Write $\delta_k = \min_{m \in \mathbb{Z}} |\theta_k - m\pi/2| \in [0, \pi/4]$. For one block the affinity is the relative area of intersection of a square and its rotation. Removing the four corner triangles gives, for $0 < \delta \leq \pi/4$,

$$\begin{aligned} A(\mu_{u,2}, (O_\delta)_\# \mu_{u,2}) &= 1 - \frac{(\cos \delta + \sin \delta - 1)^2}{2 \sin \delta \cos \delta}, \\ 1 - A(\mu_{u,2}, (O_\delta)_\# \mu_{u,2}) &= \frac{\delta}{2} + O(\delta^2). \end{aligned}$$

At $\delta = 0$, the affinity is one. Each nonzero δ leaves positive area on both sides of the support difference. Consequently [theorem 5.4](#) gives the complete trichotomy

$$\begin{array}{l|l} \delta_k = 0 \text{ for every } k & \nu_T = \mu \\ 0 < \sum_k \delta_k < \infty & \nu_T \not\ll \mu, \quad \nu_T \not\ll \mu, \quad \mu \not\ll \nu_T \\ \sum_k \delta_k = \infty & \nu_T \perp \mu. \end{array}$$

In particular, $\theta_k = 1/k$ gives $T - I \in \mathcal{S}_2$ but $\nu_T \perp \mu$. With $\theta_k = 1/k^2$, the laws instead have positive affinity and neither is absolutely continuous with respect to the other.

For comparison, let f be the variance-one rescaling of $(1 - x^2)^r \mathbf{1}_{\{|x| < 1\}}$, $r > 1$, put $\mu_{f,2} = (f(x) dx)^{\otimes 2}$, and set $J = 4\|(\sqrt{f})'\|_2^2 > 1$. The skew-generator formula gives

$$1 - A(\mu_{f,2}, (O_\theta)_\# \mu_{f,2}) = \frac{J - 1}{4} \theta^2 + o(\theta^2). \quad (91)$$

Only signed-permutation rotations preserve the square support, but every rotation retains positive overlap near the origin. Continuity and (91) therefore replace $\sum_k \delta_k$ by $\sum_k \delta_k^2$ in the trichotomy, while equivalence still requires every $\delta_k = 0$.

5.3 Complete linear classification for bounded and half-line laws

Let X be centered with variance one, put $\rho = \text{Law}(X)$, $\rho^- = \text{Law}(-X)$, and $\mu = \rho^{\otimes \mathbb{N}}$, and use the row-series action of [lemma 3.1](#). Conditional expectations turn a support boundary into an ℓ^1 row constraint; the inverse relations then force a monomial operator, without density or Fisher assumptions.

Theorem 5.3 (Linear equivalence for bounded and half-line coordinate laws). *Let $T \in \text{GL}(H)$.*

1. *Suppose $a = \text{ess inf } X < 0 < b = \text{ess sup } X$ are finite. Then $\nu_T \sim \mu$ forces T to be a signed permutation. The complete equivalence group is as follows:*

<i>relation between ρ^- and ρ</i>	<i>allowed coordinate signs</i>
$\rho^- = \rho$	<i>all signs</i>
$\rho^- \sim \rho, \rho^- \neq \rho$	<i>only finitely many negative signs</i>
$\rho^- \not\sim \rho$	<i>no negative signs.</i>

In particular, $a + b \neq 0$ permits only ordinary permutations. The support need not fill an interval, and a density is not required.

2. *Suppose $\text{ess inf } X = a > -\infty$ and $\text{ess sup } X = +\infty$. Then*

$$\nu_T \sim \mu \iff T \text{ is an ordinary coordinate permutation.}$$

The same conclusion holds for a law bounded above and unbounded below.

In the second case every equivalence is exact invariance. Failure of any of these equivalence criteria does not, by itself, imply singularity.

Proof. Assume $\nu_T \sim \mu$, and set $S = T^{-1}$. The measurable inverse action also gives $\nu_S \sim \mu$. For every row v of T or S , the series $Z = \sum_j v_j X_j$ converges in L^2 , and

$$\mathbb{E}[Z \mid X_1, \dots, X_m] = \sum_{j \leq m} v_j X_j. \quad (92)$$

In the bounded case $Z \in [a, b]$ almost surely, so its conditional expectation does too. The essential width of the finite sum in (92) is $(b - a) \sum_{j \leq m} |v_j|$; hence every row of T and S has ℓ^1 -norm at most one.

Let $A_{ij} = |T_{ij}|$ and $B_{ij} = |S_{ij}|$. The ℓ^2 row/column Cauchy–Schwarz inequality makes the entries of TS and ST absolutely convergent, so $AB \geq I$ and $BA \geq I$ entrywise. Tonelli and the row bounds give

$$\sum_j (AB)_{ij} = \sum_k A_{ik} \sum_j B_{kj} \leq 1, \quad \sum_j (BA)_{ij} \leq 1.$$

Thus $AB = BA = I$. For each i , choose k with $A_{ik} B_{ki} > 0$. The off-diagonal entries of AB force row k of B to vanish outside column i , and those of BA force row i of A to vanish outside column k . This gives a bijective monomial pattern. Its weights and reciprocals are at most one by the row bounds, so every weight is one; hence T is a signed permutation.

For a negative row sign, scalar equivalence requires $\rho^- \sim \rho$, and therefore $a = -b$. Each negative sign contributes $A(\rho, \rho^-)$ to the product affinity. This number equals one exactly when $\rho^- = \rho$; otherwise Kakutani’s theorem [17] permits only finitely many negative signs. This proves the three alternatives.

In the lower-bounded, upper-unbounded case, centering and nondegeneracy give $a < 0$. Every finite sum in (92) is bounded below by a , so no coefficient can be negative. Its essential infimum is $a \sum_{j \leq m} v_j$, whence

$$a \sum_{j \leq m} v_j \geq a, \quad \text{hence} \quad \sum_j v_j \leq 1.$$

Thus T and S are nonnegative with row sums at most one, and the preceding inverse-matrix argument makes T an ordinary permutation. Such permutations preserve μ ; reflection handles the upper-bounded case. $\square$

If Y has a Gamma law with shape $r_\Gamma > 0$ and arbitrary positive scale, its standardization has endpoints $-\sqrt{r_\Gamma}$ and $+\infty$; hence only coordinate permutations are allowed for every positive shape and scale. For compact support, signs are governed by ρ versus ρ^- : if $\rho^- \sim \rho$ but $\rho^- \neq \rho$, only finitely many negative signs are allowed, whereas unequal endpoint magnitudes, as for a centered asymmetric Beta law, preclude them.

Necessity of the inverse zero-set row condition. Even a rank-one perturbation can have positive Hellinger affinity, with $\nu_T \ll \mu$ but $\mu \not\ll \nu_T$.

Let f be the rescaling to variance one of $C(1-x^2)^2 \mathbf{1}_{\{|x|<1\}}$, and write its support as $[-a, a]$. Set $\rho(dx) = f(x) dx$. Its zero-extended square root belongs to $W^{1,2}(\mathbb{R})$, and $x(\sqrt{f})' \in L^2(\mathbb{R})$. For $0 < c < 1$, set

$$T = c \oplus I, \quad f_c(y) = c^{-1} f(y/c), \quad \mu = \rho^{\otimes \mathbb{N}}.$$

Then $T - I$ has rank one and

$$\nu_T = (f_c(y) dy) \otimes \rho^{\otimes \{2,3,\dots\}} \ll \mu, \quad \mu \not\ll \nu_T.$$

Indeed, f_c is positive on $(-ca, ca)$ and zero on the rest of $(-a, a)$. Nevertheless,

$$A(\mu, \nu_T) = \int \sqrt{f(y)f_c(y)} dy > 0, \quad \sup_n E_n(T) < \infty.$$

Thus neither Hilbert–Schmidt proximity nor bounded projected Hellinger energy implies equivalence without the inverse zero-set row condition. The failure occurs with finite location and scale Fisher information.

5.4 Block and finite-marginal criteria

The boundary examples require a block criterion separating local absolute continuity from accumulated Hellinger loss.

Theorem 5.4 (Block products without equivalence of the factors). *Let $\mu = \bigotimes_{k \geq 1} \lambda_k$ and $\nu = \bigotimes_{k \geq 1} \eta_k$ be probability products over finite-dimensional coordinate blocks. Set*

$$r_k = A(\lambda_k, \eta_k), \quad L = \sum_k -\log r_k \in [0, \infty], \quad -\log 0 = +\infty.$$

Then

$$A(\mu, \nu) = \prod_k r_k, \quad \mu \perp \nu \iff L = \infty. \quad (93)$$

If $L < \infty$, then

$$\begin{aligned} \nu \ll \mu &\iff \eta_k \ll \lambda_k \text{ for every } k, \\ \mu \ll \nu &\iff \lambda_k \ll \eta_k \text{ for every } k. \end{aligned}$$

Thus positive affinity is compatible with equivalence, with exactly one of $\nu \ll \mu$ and $\mu \ll \nu$, or with neither. For a linear block $A_k \in \text{GL}(d_k)$ and $\lambda_k = \rho^{\otimes d_k}$, where $\rho(dx) = f(x) dx$, the comparison density is

$$|\det A_k|^{-1} \prod_{i=1}^{d_k} f((A_k^{-1}y)_i).$$

The local absolute-continuity conditions concern the positivity set of this density modulo null sets, not merely the closure of its support. If $L < \infty$, take $m_k = (\lambda_k + \eta_k)/2$ with densities p_k, q_k . Then the masses of the parts of ν absolutely continuous relative to μ , and of μ relative to ν , are respectively

$$\prod_k \eta_k(\{p_k > 0\}), \quad \prod_k \lambda_k(\{q_k > 0\}). \quad (94)$$

Proof. Affinity tensorizes on finite block prefixes, and projected affinities decrease to global affinity by the conditional-expectation argument in [proposition 5.6](#). This proves (93).

The local absolute-continuity conditions are necessary by projection. If $\eta_k \ll \lambda_k$ for every k and $\prod_k r_k > 0$, set $\ell_n = \prod_{k \leq n} d\eta_k/d\lambda_k$. Independence gives

$$\|\sqrt{\ell_m} - \sqrt{\ell_n}\|_{L^2(\mu)}^2 = 2 \left(1 - \prod_{n < k \leq m} r_k \right) \longrightarrow 0.$$

Their L^2 limit has norm one, and its square defines a probability measure which cylinder tests identify with ν ; hence $\nu \ll \mu$. Interchanging μ and ν proves the reverse assertion. The local condition cannot be omitted [17, 32]. For the mass formula, restrict to $C = \prod_k \{p_k q_k > 0\}$. Cauchy–Schwarz and $\prod_k r_k > 0$ make the conditional affinities have positive product, so the conditional product laws are equivalent. Their restrictions to C are precisely the two absolutely continuous parts, with the masses in (94). $\square$

Gamma scale changes and the effect of centering. For the uncentered shape- r_Γ , scale-one Gamma law, direct integration gives

$$\int_0^\infty \sqrt{f(y)c^{-1}f(y/c)} dy = \left(\frac{2\sqrt{c}}{1+c} \right)^{r_\Gamma} = \cosh\left(\frac{\log c}{2}\right)^{-r_\Gamma}.$$

Since $\log \cosh(u/2) \sim u^2/8$, Kakutani gives

$$\bigotimes_k \text{Law}(c_k Y) \sim \bigotimes_k \text{Law}(Y) \iff \sum_k (\log c_k)^2 < \infty, \quad c_k > 0. \quad (95)$$

Equivalently, $\sum_k (c_k - 1)^2 < \infty$. For $X = (Y - r_\Gamma)/\sqrt{r_\Gamma}$, the dilation $Y \mapsto cY$ becomes

$$X \mapsto cX + \sqrt{r_\Gamma}(c - 1),$$

rather than the homogeneous map $X \mapsto cX$.

Corollary 5.5 (Coordinatewise affine equivalence for Gamma products). *Let μ be the product of the shape- r_Γ Gamma law centered with variance one. For $T \in \text{GL}(H)$ and $b \in \ell^2$, the law of $\Phi_T(x) + b$ under μ is equivalent to μ if and only if there are a coordinate permutation π and numbers $c_i > 0$ such that*

$$(Tx)_i = c_i x_{\pi(i)}, \quad b_i = \sqrt{r_\Gamma}(c_i - 1), \quad \sum_i (\log c_i)^2 < \infty.$$

Proof. Set $S = T^{-1}$, so $Sb \in \ell^2$. Cauchy–Schwarz gives $\Phi_S(z + b) = \Phi_S(z) + Sb$ wherever the row limits exist, so the inverse is $z \mapsto \Phi_S(z) - Sb$. Translation by $d \in \ell^2$ changes a covariance constant C_λ to at most $2(C_\lambda + \|d\|_2^2)$, and a bounded linear pushforward by R changes it to at

most $C_\lambda \|R\|_{\text{op}}^2$. Thus every affine-orbit law satisfies the hypothesis of [lemma 3.1](#); that lemma and the translation identity give the map and its inverse a common full-measure carrier. Conditioning finite row sums as in the proof of [theorem 5.3](#) shows that both T and T^{-1} have nonnegative entries: a negative coefficient would create an unbounded lower tail that no fixed shift can remove. The inverse identities then make T monomial with positive weights c_i . Equality of the scalar lower endpoints forces $-\sqrt{r_\Gamma}c_i + b_i = -\sqrt{r_\Gamma}$. Equation (95) completes both directions. $\square$

Finite-marginal criterion.

Proposition 5.6 (Finite-marginal tails characterize equivalence). *Let μ, ν be any probability measures on $\mathbb{R}^\mathbb{N}$, and let $\mu^{(n)}, \nu^{(n)}$ be their first n marginals. Choose densities p_n, q_n with respect to any common dominating measure m_n . For $M > 1$, define*

$$\text{Tail}_n(M) = \int_{\{q_n > Mp_n\}} q_n dm_n + \int_{\{p_n > Mq_n\}} p_n dm_n.$$

Then

$$\mu \sim \nu \iff \lim_{M \rightarrow \infty} \sup_n \text{Tail}_n(M) = 0. \quad (96)$$

Equivalently, every finite pair is equivalent and its forward and reverse likelihood-ratio martingales are uniformly integrable in their respective reference measures. Without any finite-marginal equivalence assumption,

$$A(\mu^{(n)}, \nu^{(n)}) \downarrow A(\mu, \nu), \quad \sup_n [-2 \log A(\mu^{(n)}, \nu^{(n)})] < \infty \iff \mu \not\sim \nu.$$

Proof. If the limit in (96) is zero, a set where $p_n = 0 < q_n$ must have zero $\nu^{(n)}$ -mass for every n ; the second integral gives the reverse, so $\mu^{(n)} \sim \nu^{(n)}$. With $\ell_n(x) = q_n(\pi_n x)/p_n(\pi_n x)$, the two summands in $\text{Tail}_n(M)$ are

$$\mathbb{E}_\mu[\ell_n \mathbf{1}_{\{\ell_n > M\}}], \quad \mathbb{E}_\nu[\ell_n^{-1} \mathbf{1}_{\{\ell_n^{-1} > M\}}].$$

Thus the limit is exactly uniform integrability of the two mean-one likelihood-ratio martingales. Their L^1 limits have mass one, and cylinder tests prove mutual absolute continuity. Conversely, under equivalence they are conditional expectations of the full likelihood ratio and its reciprocal, hence uniformly integrable.

For affinity, take $m = (\mu + \nu)/2$, write $u = d\mu/dm$, $v = d\nu/dm$, and let u_n, v_n be their conditional expectations on the first n coordinates. Since $u, v \leq 2$, bounded martingale convergence gives

$$A(\mu^{(n)}, \nu^{(n)}) = \int \sqrt{u_n v_n} dm \longrightarrow \int \sqrt{uv} dm.$$

Conditional Cauchy–Schwarz gives monotonicity, and the limit vanishes exactly for mutually singular measures. $\square$

Roles of the finite-variance hypotheses. The positive-density theorem includes variance-one rescalings of the Laplace and e^{-x^4} laws, and the centered, variance-one version of the law with density before normalization proportional to

$$(1 + y^2)^{-(\tau+1)/2} e^{\vartheta \arctan y}, \quad \tau > 2, \quad \vartheta \in \mathbb{R}.$$

Its pre-normalization score is $((\tau + 1)y - \vartheta)/(1 + y^2)$; it and its product with y are bounded, and these bounds persist under normalization. For $2 < \tau \leq 4$ the fourth moment is infinite, while $\vartheta \neq 0$ gives unequal tail constants. Thus neither symmetry nor a fourth moment is assumed.

When f has zeros, the forward and inverse zero-set row conditions determine whether row laws avoid them. The second moment makes ℓ^2 row series converge, their squares uniformly integrable, and the Gaussian-summand extraction in [lemma 3.8](#) possible. By contrast, [lemma 2.1](#) assumes only a first moment and has broader scope than [theorem 1.1](#).

Location and scale Fisher information control dimension-free Hellinger path and generator bounds. The tests H_A require no higher moments and pair those bounds with every matrix tangent. In the necessity proof, [lemma 3.9](#) makes remote swaps asymptotically invisible in Hellinger distance, avoiding variance bounds under conjugated output laws. Thus the orthogonal theorem drops scale Fisher information, while the full bounded invertible theorem retains it.

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